

QUANTUM MECHANICS — COMPREHENSIVE STUDY COMPENDIUM

Chapter 2: The Time-Independent Schrödinger Equation

Complete Examples & Problems with In-Depth Step-by-Step Solutions & Mathematical Foundations

Textbook: D. J. Griffiths, *Introduction to Quantum Mechanics* **Coverage:** Sections 2.1 – 2.6 & Further Problems

Contents: 6 Examples & 64 Problems with Complete Derivations

SECTION 2.1

Stationary States

Separation of variables, time-independent Schrödinger equation, properties of stationary states, and energy expectation values.

EXAMPLE 2.1 Superposition of Two Stationary States

Suppose a particle starts out in a linear combination of just two stationary states:

$$\Psi(x, 0) = c_1\psi_1(x) + c_2\psi_2(x).$$

(To keep things simple assume that the constants c_n and the states $\psi_n(x)$ are real.) What is the wave function $\Psi(x, t)$ at subsequent times? Find the probability density $|\Psi(x, t)|^2$, and describe its motion.

DETAILED STEP-BY-STEP SOLUTION**Step 1: Construct the full time-dependent wave function $\Psi(x, t)$.**

The time-dependent Schrödinger equation is linear. For any stationary state $\psi_n(x)$ satisfying $\hat{H}\psi_n = E_n\psi_n$, its time evolution is given by multiplying by the phase factor $e^{-iE_nt/\hbar}$. Therefore, the linear combination evolves as:

$$\Psi(x, t) = c_1\psi_1(x)e^{-iE_1t/\hbar} + c_2\psi_2(x)e^{-iE_2t/\hbar}.$$

Step 2: Calculate the probability density $|\Psi(x, t)|^2 = \Psi^*(x, t)\Psi(x, t)$ step by step.

Since $c_1, c_2, \psi_1(x)$, and $\psi_2(x)$ are real, the complex conjugate is $\Psi^*(x, t) = c_1\psi_1(x)e^{iE_1t/\hbar} + c_2\psi_2(x)e^{iE_2t/\hbar}$.

Multiplying out the terms using the **aligned** multi-line environment:

$$\begin{aligned} |\Psi(x, t)|^2 &= \left(c_1\psi_1e^{iE_1t/\hbar} + c_2\psi_2e^{iE_2t/\hbar} \right) \left(c_1\psi_1e^{-iE_1t/\hbar} + c_2\psi_2e^{-iE_2t/\hbar} \right) \\ &= c_1^2\psi_1^2e^{i(E_1-E_1)t/\hbar} + c_2^2\psi_2^2e^{i(E_2-E_2)t/\hbar} + c_1c_2\psi_1\psi_2e^{i(E_1-E_2)t/\hbar} + c_1c_2\psi_1\psi_2e^{-i(E_2-E_1)t/\hbar} \\ &= c_1^2\psi_1(x)^2 + c_2^2\psi_2(x)^2 + c_1c_2\psi_1(x)\psi_2(x) \left[e^{-i(E_2-E_1)t/\hbar} + e^{i(E_2-E_1)t/\hbar} \right] \\ &= c_1^2\psi_1(x)^2 + c_2^2\psi_2(x)^2 + 2c_1c_2\psi_1(x)\psi_2(x) \cos\left(\frac{E_2-E_1}{\hbar}t\right). \end{aligned}$$

Step 3: Physical interpretation of the motion.

While an individual stationary state has a time-independent probability density ($|\Psi_n|^2 = |\psi_n|^2$), a superposition of two stationary states has an interference cross-term that oscillates sinusoidally in time with angular frequency:

$$\omega = \frac{E_2 - E_1}{\hbar}.$$

The probability density sloshes back and forth across the potential well at frequency ω .

FINAL RESULT / CONCLUSION:

$$\Psi(x, t) = c_1\psi_1(x)e^{-iE_1t/\hbar} + c_2\psi_2(x)e^{-iE_2t/\hbar} \text{ and } |\Psi(x, t)|^2 = c_1^2\psi_1^2 + c_2^2\psi_2^2 + 2c_1c_2\psi_1\psi_2 \cos\left(\frac{E_2-E_1}{\hbar}t\right), \text{ oscillating with frequency } \omega = \frac{E_2-E_1}{\hbar}.$$

PROBLEM 2.1 General Theorems for Normalizable Stationary States ★

Prove the following three theorems:

- (a) For normalizable solutions, the separation constant E must be real. *Hint:* Write E as $E_0 + i\Gamma$ (with E_0 and Γ real), and show that if conservation of probability $\frac{d}{dt} \int |\Psi|^2 dx = 0$ is to hold for all t , Γ must be zero.
- (b) The time-independent wave function $\psi(x)$ can always be taken to be real (unlike $\Psi(x, t)$, which is necessarily complex). This doesn't mean that every solution to the time-independent Schrödinger equation is real; what it says is that if you've got one that is not, it can always be expressed as a linear combination of solutions (with the same energy) that are. So you might as well stick to ψ s that are real. *Hint:* If $\psi(x)$ satisfies $\hat{H}\psi = E\psi$, for a given E , so too does its complex conjugate ψ^* , and hence also the real linear combinations $(\psi + \psi^*)$ and $i(\psi - \psi^*)$.
- (c) If $V(x)$ is an even function (that is, $V(-x) = V(x)$) then $\psi(x)$ can always be taken to be either even or odd. *Hint:* If $\psi(x)$ satisfies the time-independent Schrödinger equation for a given E , so too does $\psi(-x)$, and hence also the even and odd linear combinations $\psi(x) \pm \psi(-x)$.

DETAILED STEP-BY-STEP SOLUTION**Part (a): Proof that E is purely real**

Let $E = E_0 + i\Gamma$ where $E_0, \Gamma \in \mathbb{R}$.

$$\begin{aligned}\Psi(x, t) &= \psi(x)e^{-iEt/\hbar} = \psi(x)e^{-i(E_0+i\Gamma)t/\hbar} \\ &= \psi(x)e^{-iE_0t/\hbar}e^{-i(i\Gamma)t/\hbar} = \psi(x)e^{-iE_0t/\hbar}e^{\Gamma t/\hbar}.\end{aligned}$$

Now evaluate the total normalization integral at time t :

$$\int_{-\infty}^{+\infty} |\Psi(x, t)|^2 dx = e^{2\Gamma t/\hbar} \int_{-\infty}^{+\infty} |\psi(x)|^2 dx = e^{2\Gamma t/\hbar}.$$

For conservation of probability to hold for all times t , we require:

$$\frac{d}{dt} \left(e^{2\Gamma t/\hbar} \right) = \frac{2\Gamma}{\hbar} e^{2\Gamma t/\hbar} = 0 \implies \Gamma = 0.$$

Hence, the imaginary part of E must vanish, proving $E = E_0 \in \mathbb{R}$.

Part (b): Proof that $\psi(x)$ can always be taken as real

The Time-Independent Schrödinger Equation (TISE) is:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x).$$

Since $\hbar, m, V(x)$, and E are purely real, taking the complex conjugate gives:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi^*(x)}{dx^2} + V(x)\psi^*(x) = E\psi^*(x).$$

Thus, if $\psi(x)$ satisfies the TISE with energy E , then its complex conjugate $\psi^*(x)$ is also an exact solution with the same energy E . Constructing real linear combinations:

$$\psi_1(x) \equiv \frac{\psi(x) + \psi^*(x)}{2} = \text{Re}[\psi(x)], \quad \psi_2(x) \equiv \frac{\psi(x) - \psi^*(x)}{2i} = \text{Im}[\psi(x)].$$

Both $\psi_1(x)$ and $\psi_2(x)$ are real eigenfunctions with energy E .

Part (c): Proof of definite parity for symmetric potentials $V(-x) = V(x)$

Given $-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$, substitute $x \rightarrow -x$. Using $V(-x) = V(x)$:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(-x)}{dx^2} + V(x)\psi(-x) = E\psi(-x).$$

Constructing the symmetric (even) and antisymmetric (odd) superpositions:

$$\psi_{\text{even}}(x) \equiv \frac{\psi(x) + \psi(-x)}{2}, \quad \psi_{\text{odd}}(x) \equiv \frac{\psi(x) - \psi(-x)}{2}.$$

Both are valid energy eigenstates with eigenvalue E , having definite parity

$$\psi_{\text{even}}(-x) = \psi_{\text{even}}(x) \text{ and } \psi_{\text{odd}}(-x) = -\psi_{\text{odd}}(x).$$

FINAL RESULT / CONCLUSION:

Proven: (a) E must be strictly real ($\Gamma = 0$); (b) $\psi(x)$ can always be chosen real; (c) for symmetric potentials $V(-x) = V(x)$, eigenstates have definite even or odd parity.



PROBLEM 2.2 Minimum Energy Theorem ★

Show that E must exceed the minimum value of $V(x)$, for every normalizable solution to the time-independent Schrödinger equation: $-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$. What is the classical analog to this statement?

HINT / NOTE

Rewrite the time-independent Schrödinger equation in the form

$$\frac{d^2\psi}{dx^2} = \frac{2m}{\hbar^2} [V(x) - E]\psi;$$

if $E < V_{\min}$, then ψ and its second derivative always have the same sign—argue that such a function cannot be normalized.

DETAILED STEP-BY-STEP SOLUTION

Step 1: Rewrite the Schrödinger equation in curvature form.

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x)\psi = E\psi \implies \frac{d^2\psi}{dx^2} = \frac{2m}{\hbar^2} [V(x) - E]\psi(x).$$

Step 2: Sign analysis of the second derivative for $E < V_{\min}$.

Let $V_{\min}$ be the global minimum of $V(x)$, so $V(x) \geq V_{\min}$ for all $x \in (-\infty, \infty)$.

If $E < V_{\min}$, then $V(x) - E > 0$ strictly for all x . Therefore, the factor $\frac{2m}{\hbar^2} [V(x) - E]$ is always positive.

This means $\frac{d^2\psi}{dx^2}$ always has the **exact same sign** as $\psi(x)$:

- Where $\psi(x) > 0$, $\psi''(x) > 0 \implies$ the function is concave upwards (curves away from the x -axis).
- Where $\psi(x) < 0$, $\psi''(x) < 0 \implies$ the function is concave downwards (curves away from the x -axis).

Step 3: Why normalizability fails.

For $\psi(x)$ to be normalizable, we must have $\psi(x) \rightarrow 0$ as $x \rightarrow \pm\infty$.

Assume $\psi(x) > 0$ as $x \rightarrow -\infty$ with $\psi(-\infty) = 0$. Since $\psi > 0$, its slope must be positive ($\psi' > 0$) as it rises away from 0. Since $\psi'' > 0$, the slope ψ' continues to increase monotonically. For $\psi(x)$ to ever return to 0 as $x \rightarrow +\infty$, its slope would have to turn negative ($\psi' < 0$), which requires a region of downward concavity ($\psi'' < 0$). But ψ'' can never be negative while $\psi > 0$.

Thus, $\psi(x)$ diverges exponentially to $+\infty$ as $x \rightarrow +\infty$. Such a solution cannot be normalized ($\int_{-\infty}^{\infty} |\psi|^2 dx = \infty$).

Therefore, we must have $E > V_{\min}$ for every normalizable stationary state.

Step 4: Classical analog.

Classically, total mechanical energy is $E = T + V(x) = \frac{p^2}{2m} + V(x)$. Because momentum is real, kinetic energy $T = \frac{p^2}{2m} \geq 0$, which mandates $E \geq V(x) \geq V_{\min}$. A particle cannot have total energy less than the lowest potential energy in the system.

FINAL RESULT / CONCLUSION:

Proven: $E > V_{\min}$ for all normalizable states. Classical analog: kinetic energy is non-negative ($T = E - V \geq 0 \implies E \geq V_{\min}$).

SECTION 2.2

The Infinite Square Well

Stationary states $\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$, energy eigenvalues $E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$, completeness, Fourier expansion, and quantum dynamics.

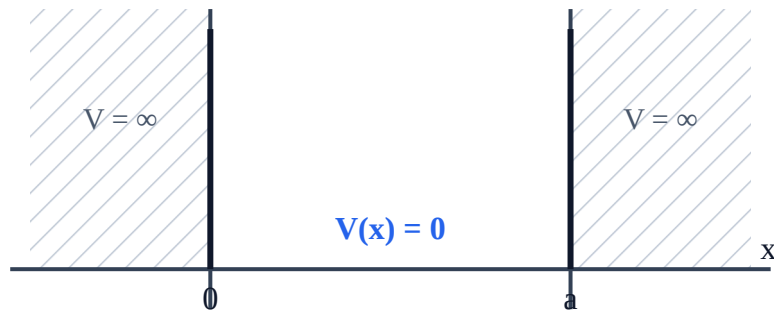


Figure 2.1: The infinite square well potential: $V(x) = 0$ for $0 \leq x \leq a$, and $V(x) = \infty$ elsewhere.

EXAMPLE 2.2 Parabolic Initial State in the Infinite Square Well

A particle in the infinite square well potential ($V(x) = 0$ for $0 \leq x \leq a$, ∞ otherwise) has the initial wave function:

$$\Psi(x, 0) = Ax(a - x), \quad (0 \leq x \leq a),$$

for some constant A (see Figure 2.3). Outside the well, $\Psi = 0$. Find $\Psi(x, t)$.

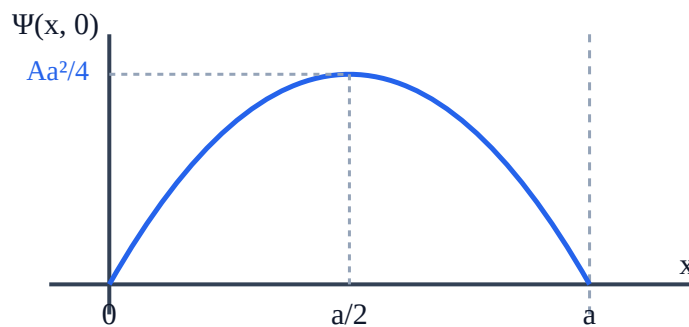


Figure 2.3: The starting wave function $\Psi(x, 0) = Ax(a - x)$ in Example 2.2.

DETAILED STEP-BY-STEP SOLUTION

Step 1: Normalization of $\Psi(x, 0) = Ax(a - x)$.

$$\begin{aligned} 1 &= \int_0^a |\Psi(x, 0)|^2 dx = A^2 \int_0^a x^2(a - x)^2 dx \\ &= A^2 \int_0^a (a^2x^2 - 2ax^3 + x^4) dx \\ &= A^2 \left[a^2 \frac{x^3}{3} - 2a \frac{x^4}{4} + \frac{x^5}{5} \right]_0^a \\ &= A^2 a^5 \left(\frac{1}{3} - \frac{1}{2} + \frac{1}{5} \right) = A^2 a^5 \left(\frac{10 - 15 + 6}{30} \right) = A^2 \frac{a^5}{30}. \end{aligned}$$

$$1 = A^2 \frac{a^5}{30} \implies A = \sqrt{\frac{30}{a^5}}.$$

Step 2: Explicit Integration for Fourier coefficients c_n .

The expansion coefficients are $c_n = \sqrt{\frac{2}{a}} \sqrt{\frac{30}{a^5}} I_n = \frac{2\sqrt{15}}{a^3} I_n$, where $I_n \equiv \int_0^a (ax - x^2) \sin(kx) dx$ with $k \equiv \frac{n\pi}{a}$.

Performing integration by parts with $u = ax - x^2$, $dv = \sin(kx) dx$:

$$\begin{aligned} I_n &= \left[-(ax - x^2) \frac{\cos(kx)}{k} \right]_0^a + \frac{1}{k} \int_0^a (a - 2x) \cos(kx) dx \\ &= 0 + \frac{1}{k} \left(\left[(a - 2x) \frac{\sin(kx)}{k} \right]_0^a - \int_0^a (-2) \frac{\sin(kx)}{k} dx \right) \\ &= \frac{2}{k^2} \left[-\frac{\cos(kx)}{k} \right]_0^a = \frac{2}{k^3} [1 - \cos(n\pi)] = \frac{2a^3}{n^3 \pi^3} [1 - (-1)^n]. \end{aligned}$$

Thus $c_n = 0$ for even n , and for odd n :

$$c_n = \frac{2\sqrt{15}}{a^3} \cdot \frac{4a^3}{n^3 \pi^3} = \frac{8\sqrt{15}}{\pi^3 n^3}.$$

Step 3: Assemble the full time-dependent wave function $\Psi(x, t)$.

$$\Psi(x, t) = \frac{16\sqrt{30}}{\pi^3 \sqrt{a}} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^3} \sin\left(\frac{n\pi x}{a}\right) \exp\left(-i \frac{n^2 \pi^2 \hbar}{2ma^2} t\right).$$

FINAL RESULT / CONCLUSION:

$$\Psi(x, t) = \frac{16\sqrt{30}}{\pi^3 \sqrt{a}} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^3} \sin\left(\frac{n\pi x}{a}\right) \exp\left(-i \frac{n^2 \pi^2 \hbar}{2ma^2} t\right).$$

EXAMPLE 2.3 Energy Probability and Expectation Value for Parabolic State

Check that the probability conservation sum $\sum_{n=1}^{\infty} |c_n|^2 = 1$ is satisfied for the wave function in [Example 2.2](#): where $\Psi(x, 0) = \sqrt{\frac{30}{a^5}} x(a - x)$ and $c_n = \frac{8\sqrt{15}}{\pi^3 n^3}$ for odd n . If you measured the energy of a particle in this state, what is the most probable result? What is the expectation value of the energy $\langle H \rangle$?

DETAILED STEP-BY-STEP SOLUTION

Step 1: Verify probability conservation $\sum_{n=1}^{\infty} |c_n|^2 = 1$.

$$\sum_{n=1,3,5,\dots} |c_n|^2 = \frac{960}{\pi^6} \sum_{n=1,3,5,\dots} \frac{1}{n^6} = \frac{960}{\pi^6} \left(\frac{\pi^6}{960} \right) = 1. \quad \checkmark$$

Step 2: Most probable energy measurement.

For ground state $n = 1$: $P(E_1) = |c_1|^2 = \frac{960}{\pi^6} \approx 0.998555$ (99.86%).

Step 3: Calculate expectation value $\langle H \rangle = \sum |c_n|^2 E_n$.

$$\begin{aligned} \langle H \rangle &= \sum_{n=1,3,5,\dots} \left(\frac{960}{\pi^6 n^6} \right) \left(\frac{n^2 \pi^2 \hbar^2}{2ma^2} \right) \\ &= \frac{480 \hbar^2}{\pi^4 ma^2} \sum_{n=1,3,5,\dots} \frac{1}{n^4} \\ &= \frac{480 \hbar^2}{\pi^4 ma^2} \left(\frac{\pi^4}{96} \right) = \frac{5 \hbar^2}{ma^2}. \end{aligned}$$

FINAL RESULT / CONCLUSION:

$$\sum |c_n|^2 = 1; \text{ most probable energy is } E_1 = \frac{\pi^2 \hbar^2}{2ma^2} \text{ (99.86\% probability); } \langle H \rangle = \frac{5 \hbar^2}{ma^2} \approx 1.013 E_1.$$

PROBLEM 2.3 Absence of Zero and Negative Energy Solutions

Show that there is no acceptable solution to the (time-independent) Schrödinger equation for the infinite square well with $E = 0$ or $E < 0$. (This is a special case of the general theorem in [Problem 2.2](#), but this time do it by explicitly solving the Schrödinger equation, and showing that you cannot satisfy the boundary conditions $\psi(0) = \psi(a) = 0$.)

DETAILED STEP-BY-STEP SOLUTION

Inside the well ($0 \leq x \leq a$), $V(x) = 0$. The TISE is $\frac{d^2\psi}{dx^2} = -\frac{2mE}{\hbar^2}\psi$.

Case 1: $E = 0$.

$$\frac{d^2\psi}{dx^2} = 0 \implies \psi(x) = Ax + B.$$

Boundary conditions: $\psi(0) = B = 0 \implies \psi(x) = Ax$. Then $\psi(a) = Aa = 0 \implies A = 0$. Thus $\psi(x) \equiv 0$ (trivial).

Case 2: $E < 0$.

Let $E = -|E|$ and $k \equiv \frac{\sqrt{2m|E|}}{\hbar} > 0$. The solution is $\psi(x) = C \sinh(kx) + D \cosh(kx)$.

Boundary conditions: $\psi(0) = D = 0 \implies \psi(x) = C \sinh(kx)$.

Then $\psi(a) = C \sinh(ka) = 0$. Since $ka > 0$, $\sinh(ka) > 0$ strictly, forcing $C = 0 \implies \psi(x) \equiv 0$.

Thus, no physical state exists for $E \leq 0$.

FINAL RESULT / CONCLUSION:

No non-trivial solution exists for $E \leq 0$ because satisfying boundary conditions $\psi(0) = \psi(a) = 0$ forces $\psi(x) = 0$ everywhere.

PROBLEM 2.4 **Uncertainties in the Infinite Square Well** ★

Calculate $\langle x \rangle$, $\langle x^2 \rangle$, $\langle p \rangle$, $\langle p^2 \rangle$, σ_x , and σ_p , for the n th stationary state of the infinite square well:

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right).$$

Check that the Heisenberg uncertainty principle $\sigma_x \sigma_p \geq \hbar/2$ is satisfied. Which state comes closest to the uncertainty limit?

DETAILED STEP-BY-STEP SOLUTION

For $\psi_n(x) = \sqrt{\frac{2}{a}} \sin(kx)$ with $k = \frac{n\pi}{a}$:

1. Position expectation values $\langle x \rangle$ and $\langle x^2 \rangle$:

$$\langle x \rangle = \frac{2}{a} \int_0^a x \sin^2(kx) dx = \frac{a}{2}.$$

$$\begin{aligned} \langle x^2 \rangle &= \frac{2}{a} \int_0^a x^2 \sin^2(kx) dx = \frac{1}{a} \int_0^a x^2 (1 - \cos(2kx)) dx \\ &= \frac{a^2}{3} - \frac{1}{a} \left(\left[x^2 \frac{\sin(2kx)}{2k} \right]_0^a - \int_0^a 2x \frac{\sin(2kx)}{2k} dx \right) \\ &= \frac{a^2}{3} - \frac{a^2}{2n^2\pi^2} = a^2 \left(\frac{1}{3} - \frac{1}{2n^2\pi^2} \right). \end{aligned}$$

$$\sigma_x = a \sqrt{\frac{1}{12} - \frac{1}{2n^2\pi^2}}.$$

2. Momentum expectation values $\langle p \rangle$ and $\langle p^2 \rangle$:

$$\langle p \rangle = 0, \quad \langle p^2 \rangle = \left(\frac{n\pi\hbar}{a} \right)^2 \implies \sigma_p = \frac{n\pi\hbar}{a}.$$

3. Uncertainty product $\sigma_x \sigma_p$:

$$\sigma_x \sigma_p = \frac{\hbar}{2} \sqrt{\frac{n^2\pi^2}{3} - 2} \geq \frac{\hbar}{2}.$$

FINAL RESULT / CONCLUSION:

$$\langle x \rangle = \frac{a}{2}, \langle x^2 \rangle = a^2 \left(\frac{1}{3} - \frac{1}{2n^2\pi^2} \right), \langle p \rangle = 0, \langle p^2 \rangle = \frac{n^2\pi^2\hbar^2}{a^2}, \sigma_x\sigma_p = \frac{\hbar}{2} \sqrt{\frac{n^2\pi^2}{3} - 2} \geq \frac{\hbar}{2}.$$

PROBLEM 2.5 Superposition of Ground and First Excited States ★

A particle in the infinite square well has as its initial wave function an even mixture of the first two stationary states:

$$\Psi(x, 0) = A[\psi_1(x) + \psi_2(x)].$$

- (a) Normalize $\Psi(x, 0)$.
- (b) Find $\Psi(x, t)$ and $|\Psi(x, t)|^2$.
- (c) Compute $\langle x \rangle$. Notice that it oscillates in time. What is the angular frequency of the oscillation? What is the amplitude of the oscillation?
- (d) Compute $\langle p \rangle$ using $\langle p \rangle = m \frac{d\langle x \rangle}{dt}$.
- (e) If you measured the energy of this particle, what values might you get, and what is the probability of getting each of them? Find the expectation value of H .

DETAILED STEP-BY-STEP SOLUTION**Part (a): Find Normalization Constant A**

$$1 = |A|^2(1 + 1) = 2|A|^2 \implies A = \frac{1}{\sqrt{2}}.$$

Part (b): Construct $\Psi(x, t)$ and $|\Psi(x, t)|^2$

$$\Psi(x, t) = \frac{1}{\sqrt{2}}\psi_1(x)e^{-iE_1t/\hbar} + \frac{1}{\sqrt{2}}\psi_2(x)e^{-iE_2t/\hbar}.$$

$$|\Psi(x, t)|^2 = \frac{1}{2}\psi_1(x)^2 + \frac{1}{2}\psi_2(x)^2 + \psi_1(x)\psi_2(x)\cos(3\omega_0t), \quad \text{where } \omega_0 \equiv \frac{\pi^2\hbar}{2ma^2}.$$

Part (c): Compute $\langle x \rangle(t)$

$$\begin{aligned}
\langle x \rangle(t) &= \int_0^a x |\Psi(x, t)|^2 dx \\
&= \frac{1}{2} \int_0^a x \psi_1^2 dx + \frac{1}{2} \int_0^a x \psi_2^2 dx + \cos(3\omega_0 t) \int_0^a x \psi_1 \psi_2 dx \\
&= \frac{a}{2} - \frac{16a}{9\pi^2} \cos(3\omega_0 t).
\end{aligned}$$

Angular frequency is $\omega = 3\omega_0 = \frac{3\pi^2\hbar}{2ma^2}$. Amplitude is $M = \frac{16a}{9\pi^2} \approx 0.18a < a/2$.

Part (d): Compute $\langle p \rangle(t)$ via Ehrenfest's Theorem

$$\begin{aligned}
\langle p \rangle(t) &= m \frac{d\langle x \rangle}{dt} = m \frac{d}{dt} \left[\frac{a}{2} - \frac{16a}{9\pi^2} \cos(3\omega_0 t) \right] \\
&= m \frac{16a}{9\pi^2} (3\omega_0) \sin(3\omega_0 t) = \frac{8\hbar}{3a} \sin \left(\frac{3\pi^2\hbar}{2ma^2} t \right).
\end{aligned}$$

Part (e): Energy Measurements & Expectation Value

$P(E_1) = 1/2$, $P(E_2) = 1/2$, and $\langle H \rangle = \frac{1}{2}(E_1 + E_2) = \frac{5\pi^2\hbar^2}{4ma^2}$.

FINAL RESULT / CONCLUSION:

(a) $A = 1/\sqrt{2}$; (b) $|\Psi|^2 = \frac{1}{2}\psi_1^2 + \frac{1}{2}\psi_2^2 + \psi_1\psi_2 \cos(3\omega_0 t)$; (c) $\langle x \rangle = \frac{a}{2} - \frac{16a}{9\pi^2} \cos(3\omega_0 t)$; (d) $\langle p \rangle = \frac{8\hbar}{3a} \sin(3\omega_0 t)$; (e) $P(E_1) = P(E_2) = 1/2$, $\langle H \rangle = \frac{5\pi^2\hbar^2}{4ma^2}$.

PROBLEM 2.6 Effect of Relative Phase

Suppose we change the relative phase of ψ_1 and ψ_2 in [Problem 2.5](#): $\Psi(x, 0) = A[\psi_1(x) + e^{i\phi}\psi_2(x)]$. Find $\Psi(x, t)$, $|\Psi(x, t)|^2$, and $\langle x \rangle$, and compare your results with what you got before. Study the special cases $\phi = \pi/2$ and $\phi = \pi$.

DETAILED STEP-BY-STEP SOLUTION

Step 1: Normalization and full wave function $\Psi(x, t)$.

$$1 = |A|^2(1 + |e^{i\phi}|^2) = 2|A|^2 \implies A = \frac{1}{\sqrt{2}}.$$

$$\Psi(x, t) = \frac{1}{\sqrt{2}}\psi_1(x)e^{-iE_1t/\hbar} + \frac{1}{\sqrt{2}}\psi_2(x)e^{i\phi}e^{-iE_2t/\hbar}.$$

Step 2: Probability density $|\Psi(x, t)|^2$ and $\langle x \rangle(t)$.

$$|\Psi(x, t)|^2 = \frac{1}{2}\psi_1(x)^2 + \frac{1}{2}\psi_2(x)^2 + \psi_1(x)\psi_2(x)\cos(\omega t - \phi), \quad \text{where } \omega \equiv \frac{3\pi^2\hbar}{2ma^2}.$$

$$\langle x \rangle(t) = \frac{a}{2} - \frac{16a}{9\pi^2}\cos(\omega t - \phi).$$

FINAL RESULT / CONCLUSION:

$\langle x \rangle(t) = \frac{a}{2} - \frac{16a}{9\pi^2}\cos(\omega t - \phi)$. Relative phase ϕ creates a temporal phase lag/lead in the physical oscillation.

PROBLEM 2.7 Symmetric Triangular Wave Function ★

A particle in the infinite square well has the initial wave function:

$$\Psi(x, 0) = \begin{cases} Ax, & 0 \leq x \leq a/2, \\ A(a - x), & a/2 \leq x \leq a. \end{cases}$$

- (a) Sketch $\Psi(x, 0)$, and determine the constant A .
- (b) Find $\Psi(x, t)$.
- (c) What is the probability that a measurement of the energy would yield the ground-state value E_1 ?
- (d) Find the expectation value of the energy, using $\langle H \rangle = \sum |c_n|^2 E_n$.

DETAILED STEP-BY-STEP SOLUTION**Part (a): Sketch and Normalization Constant A**

$$1 = 2A^2 \int_0^{a/2} x^2 dx = 2A^2 \left[\frac{x^3}{3} \right]_0^{a/2} = A^2 \frac{a^3}{12} \implies A = \sqrt{\frac{12}{a^3}} = \frac{2\sqrt{3}}{a^{3/2}}.$$

Part (b): Expansion Coefficients c_n and $\Psi(x, t)$

$$\begin{aligned} c_n &= 2A \sqrt{\frac{2}{a}} \int_0^{a/2} x \sin(kx) dx = \frac{4\sqrt{6}}{a^2} \int_0^{a/2} x \sin(kx) dx \\ &= \frac{4\sqrt{6}}{a^2} \left(\left[-x \frac{\cos(kx)}{k} \right]_0^{a/2} + \frac{1}{k} \int_0^{a/2} \cos(kx) dx \right) \\ &= \frac{4\sqrt{6}}{\pi^2 n^2} (-1)^{(n-1)/2} \quad (n = 1, 3, 5, \dots). \end{aligned}$$

$$\Psi(x, t) = \frac{8\sqrt{3}}{\pi^2 \sqrt{a}} \sum_{n=1,3,5,\dots}^{\infty} \frac{(-1)^{(n-1)/2}}{n^2} \sin\left(\frac{n\pi x}{a}\right) \exp\left(-i \frac{n^2 \pi^2 \hbar}{2ma^2} t\right).$$

Part (c): Probability of Ground State $P(E_1)$

$$P(E_1) = |c_1|^2 = \frac{96}{\pi^4} \approx 0.9855 \quad (98.55\%).$$

Part (d): Energy Expectation Value $\langle H \rangle$

$$\langle H \rangle = \sum_{n=1,3,5,\dots} |c_n|^2 E_n = \frac{48\hbar^2}{\pi^2 m a^2} \left(\frac{\pi^2}{8} \right) = \frac{6\hbar^2}{m a^2}.$$

FINAL RESULT / CONCLUSION:

(a) $A = \sqrt{12/a^3}$; (b) $c_n = \frac{4\sqrt{6}}{\pi^2 n^2} (-1)^{(n-1)/2}$ for odd n (0 for even); (c) $P(E_1) = \frac{96}{\pi^4} \approx 98.55\%$; (d) $\langle H \rangle = \frac{6\hbar^2}{m a^2}$.

PROBLEM 2.8 Step-like Localization in Left Half of the Well

A particle of mass m in the infinite square well starts out in the state $\Psi(x, 0) = A$ for $0 \leq x \leq a/2$ (0 elsewhere). What is the probability that a measurement of the energy yields $E_1 = \frac{\pi^2 \hbar^2}{2ma^2}$?

DETAILED STEP-BY-STEP SOLUTION

Step 1: Normalize $\Psi(x, 0)$.

$$1 = \int_0^{a/2} |A|^2 dx = |A|^2 \frac{a}{2} \implies A = \sqrt{\frac{2}{a}}.$$

Step 2: Explicit integration for c_1 .

$$\begin{aligned} c_1 &= \int_0^{a/2} \left(\sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right) \right) \left(\sqrt{\frac{2}{a}} \right) dx = \frac{2}{a} \int_0^{a/2} \sin\left(\frac{\pi x}{a}\right) dx \\ &= \frac{2}{a} \left[-\frac{a}{\pi} \cos\left(\frac{\pi x}{a}\right) \right]_0^{a/2} = \frac{2}{\pi}. \end{aligned}$$

Step 3: Calculate energy probability $P(E_1)$.

$$P(E_1) = |c_1|^2 = \left(\frac{2}{\pi}\right)^2 = \frac{4}{\pi^2} \approx 0.4053 \quad (40.53\%).$$

FINAL RESULT / CONCLUSION:

$$P(E_1) = |c_1|^2 = \frac{4}{\pi^2} \approx 40.53\%.$$

PROBLEM 2.9 Expectation Value of Hamiltonian via Operator Integration

For the wave function in [Example 2.2](#) ($\Psi(x, 0) = \sqrt{\frac{30}{a^5}} x(a - x)$), find the expectation value of H , at time $t = 0$, the “old fashioned” way:

$$\langle H \rangle = \int \Psi(x, 0)^* \hat{H} \Psi(x, 0) dx.$$

Compare the result we got in [Example 2.3](#) ($\langle H \rangle = \frac{5\hbar^2}{ma^2}$).

DETAILED STEP-BY-STEP SOLUTION

Step 1: Compute $\hat{H}\Psi(x, 0)$ directly by differentiation.

From [Example 2.2](#), $\Psi(x, 0) = \sqrt{\frac{30}{a^5}}(ax - x^2)$.

$$\frac{d^2\Psi}{dx^2} = -2\sqrt{\frac{30}{a^5}} \Rightarrow \hat{H}\Psi(x, 0) = -\frac{\hbar^2}{2m} \left(-2\sqrt{\frac{30}{a^5}} \right) = \frac{\hbar^2}{m} \sqrt{\frac{30}{a^5}}.$$

Step 2: Evaluate the expectation value integral $\langle H \rangle = \int_0^a \Psi^* \hat{H} \Psi dx$.

$$\begin{aligned} \langle H \rangle &= \int_0^a \left[\sqrt{\frac{30}{a^5}}(ax - x^2) \right] \left[\frac{\hbar^2}{m} \sqrt{\frac{30}{a^5}} \right] dx \\ &= \frac{30\hbar^2}{ma^5} \int_0^a (ax - x^2) dx = \frac{30\hbar^2}{ma^5} \left(\frac{a^3}{6} \right) = \frac{5\hbar^2}{ma^2}. \end{aligned}$$

This matches [Example 2.3](#) exactly.

FINAL RESULT / CONCLUSION:

$$\langle H \rangle = \int_0^a \Psi(x, 0)^* \hat{H} \Psi(x, 0) dx = \frac{5\hbar^2}{ma^2}, \text{ matching Example 2.3.}$$

SECTION 2.3**The Harmonic Oscillator**

Algebraic method with ladder operators ($\hat{a}_+$, $\hat{a}_-$), analytic series method with Hermite polynomials $H_n(\xi)$, and Gaussian ground state wave functions.

EXAMPLE 2.4 Expectation Values via Ladder Operators

Find the expectation values of $\hat{x}^2$ and $\hat{p}^2$ in the n th stationary state of the harmonic oscillator using the ladder operator method, and check the uncertainty principle.

DETAILED STEP-BY-STEP SOLUTION

Step 1: Write down the ladder operator eigenvalue relation.

$$\hat{a}_+|n\rangle = \sqrt{n+1}|n+1\rangle, \quad \hat{a}_-|n\rangle = \sqrt{n}|n-1\rangle.$$

Step 2: Express position $\hat{x}$ and momentum $\hat{p}$ in terms of ladder operators.

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a}_+ + \hat{a}_-), \quad \hat{p} = i\sqrt{\frac{\hbar m\omega}{2}}(\hat{a}_+ - \hat{a}_-).$$

Step 3: Calculate expectation value $\langle x \rangle = \langle n|\hat{x}|n\rangle$.

$$\langle x \rangle = \sqrt{\frac{\hbar}{2m\omega}}\langle n|(\hat{a}_+ + \hat{a}_-)|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}[\sqrt{n+1}\langle n|n+1\rangle + \sqrt{n}\langle n|n-1\rangle] = 0.$$

Step 4: Calculate expectation value $\langle x^2 \rangle = \langle n|\hat{x}^2|n\rangle$.

$$\begin{aligned} \langle x^2 \rangle &= \frac{\hbar}{2m\omega}\langle n|(\hat{a}_+^2 + \hat{a}_+\hat{a}_- + \hat{a}_-\hat{a}_+ + \hat{a}_-^2)|n\rangle \\ &= \frac{\hbar}{2m\omega}(0 + n + (n+1) + 0) = \left(n + \frac{1}{2}\right) \frac{\hbar}{m\omega}. \end{aligned}$$

Step 5: Calculate expectation value $\langle p^2 \rangle = \langle n|\hat{p}^2|n\rangle$.

$$\begin{aligned} \langle p^2 \rangle &= -\frac{\hbar m\omega}{2}\langle n|(\hat{a}_+^2 - \hat{a}_+\hat{a}_- - \hat{a}_-\hat{a}_+ + \hat{a}_-^2)|n\rangle \\ &= -\frac{\hbar m\omega}{2}(0 - n - (n+1) + 0) = \left(n + \frac{1}{2}\right) \hbar m\omega. \end{aligned}$$

Step 6: Verify the uncertainty principle.

$$\sigma_x \sigma_p = \sqrt{\left(n + \frac{1}{2}\right)^2 \hbar^2} = \left(n + \frac{1}{2}\right) \hbar \geq \frac{\hbar}{2}.$$

FINAL RESULT / CONCLUSION:

$$\langle x^2 \rangle = (n + 1/2) \frac{\hbar}{m\omega}, \langle p^2 \rangle = (n + 1/2) \hbar m\omega, \text{ and } \sigma_x \sigma_p = (n + 1/2) \hbar \geq \hbar/2.$$

EXAMPLE 2.5 Harmonic Oscillator State Expansion and Dynamics

A particle in the harmonic oscillator potential starts out in the state $\Psi(x, 0) = A(1 - 2\sqrt{m\omega/\hbar}x)^2 e^{-\frac{m\omega}{2\hbar}x^2}$. Find A , express $\Psi(x, 0)$ as a linear combination of stationary states $\psi_n(x)$, find $\Psi(x, t)$, and determine $\langle H \rangle$.

DETAILED STEP-BY-STEP SOLUTION

Step 1: Normalization of $\Psi(x, 0) = A(1 - 2\xi)^2 e^{-\xi^2/2}$.

Let $\xi \equiv \sqrt{\frac{m\omega}{\hbar}}x$. Expand $(1 - 2\xi)^2 = 1 - 4\xi + 4\xi^2 = H_2(\xi) - 2H_1(\xi) + 3H_0(\xi)$.

In terms of stationary states $\psi_n(x)$:

$$\Psi(x, 0) = A \left(\frac{\pi\hbar}{m\omega} \right)^{1/4} [3\psi_0 - 2\sqrt{2}\psi_1 + 2\sqrt{2}\psi_2].$$

$$1 = |A|^2 \left(\frac{\pi\hbar}{m\omega} \right)^{1/2} [3^2 + (-2\sqrt{2})^2 + (2\sqrt{2})^2] = 25|A|^2 \left(\frac{\pi\hbar}{m\omega} \right)^{1/2} \implies A = \frac{1}{5} \left(\frac{m\omega}{\pi\hbar} \right)^{1/4}$$

Thus: $c_0 = \frac{3}{5}, c_1 = -\frac{2\sqrt{2}}{5}, c_2 = \frac{2\sqrt{2}}{5}$.

Step 2: Construct $\Psi(x, t)$ and $\langle H \rangle$.

$$\Psi(x, t) = \frac{3}{5}\psi_0(x)e^{-iE_0t/\hbar} - \frac{2\sqrt{2}}{5}\psi_1(x)e^{-iE_1t/\hbar} + \frac{2\sqrt{2}}{5}\psi_2(x)e^{-iE_2t/\hbar}.$$

$$\begin{aligned} \langle H \rangle &= |c_0|^2 E_0 + |c_1|^2 E_1 + |c_2|^2 E_2 \\ &= \left(\frac{9}{25} \right) \left(\frac{1}{2} \hbar\omega \right) + \left(\frac{8}{25} \right) \left(\frac{3}{2} \hbar\omega \right) + \left(\frac{8}{25} \right) \left(\frac{5}{2} \hbar\omega \right) \\ &= \frac{9 + 24 + 40}{50} \hbar\omega = \frac{73}{50} \hbar\omega = 1.46 \hbar\omega. \end{aligned}$$

FINAL RESULT / CONCLUSION:

$$c_0 = \frac{3}{5}, c_1 = -\frac{2\sqrt{2}}{5}, c_2 = \frac{2\sqrt{2}}{5}; \langle H \rangle = 1.46 \hbar\omega.$$

PROBLEM 2.10 Commutation Relations and Raising Property ★

Prove that $[\hat{a}_-, \hat{a}_+] = 1$, and that if ψ satisfies the time-independent Schrödinger equation with energy E , then $\hat{a}_+\psi$ satisfies it with energy $E + \hbar\omega$.

DETAILED STEP-BY-STEP SOLUTION

Step 1: Compute Commutator $[\hat{a}_-, \hat{a}_+]$.

$$\begin{aligned}\hat{a}_-\hat{a}_+ &= \frac{1}{2\hbar m\omega}(i\hat{p} + m\omega\hat{x})(-i\hat{p} + m\omega\hat{x}) \\ &= \frac{1}{2\hbar m\omega}(\hat{p}^2 + m^2\omega^2\hat{x}^2 + im\omega[\hat{x}, \hat{p}]) \\ &= \frac{\hat{H}}{\hbar\omega} + \frac{i}{2\hbar}(i\hbar) = \frac{\hat{H}}{\hbar\omega} - \frac{1}{2}.\end{aligned}$$

Similarly $\hat{a}_+\hat{a}_- = \frac{\hat{H}}{\hbar\omega} + \frac{1}{2}$. Subtracting gives $[\hat{a}_-, \hat{a}_+] = 1$.

Step 2: Proof of raising property $\hat{H}(\hat{a}_+\psi) = (E + \hbar\omega)(\hat{a}_+\psi)$.

$$\hat{H}\hat{a}_+ = \hbar\omega\left(\hat{a}_+\hat{a}_- + \frac{1}{2}\right)\hat{a}_+ = \hbar\omega\hat{a}_+\left(\hat{a}_-\hat{a}_+ + \frac{1}{2}\right) = \hat{a}_+(\hat{H} + \hbar\omega).$$

Applying to ψ gives $\hat{H}(\hat{a}_+\psi) = (E + \hbar\omega)(\hat{a}_+\psi)$.

FINAL RESULT / CONCLUSION:

Proven: $[\hat{a}_-, \hat{a}_+] = 1$ and $\hat{H}(\hat{a}_+\psi) = (E + \hbar\omega)(\hat{a}_+\psi)$.

PROBLEM 2.11 Constructing the First Three Harmonic Oscillator States

Find $\psi_0(x)$ by solving $\hat{a}_-\psi_0 = 0$, and normalize it. Then apply the raising operator $\hat{a}_+$ to obtain $\psi_1(x)$ and $\psi_2(x)$. Normalize each state.

DETAILED STEP-BY-STEP SOLUTION

Step 1: Ground state $\psi_0(x)$ from $\hat{a}_-\psi_0 = 0$.

$$\frac{d\psi_0}{dx} = -\frac{m\omega}{\hbar}x\psi_0 \implies \psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}.$$

Step 2: Generate $\psi_1(x) = \hat{a}_+\psi_0(x)$.

$$\psi_1(x) = \frac{1}{\sqrt{2\hbar m\omega}} \left(-\hbar \frac{d}{dx} + m\omega x\right) \psi_0 = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \sqrt{\frac{2m\omega}{\hbar}} x e^{-\frac{m\omega}{2\hbar}x^2}.$$

Step 3: Generate $\psi_2(x) = \frac{1}{\sqrt{2}}\hat{a}_+\psi_1(x)$.

$$\psi_2(x) = \frac{1}{\sqrt{8}} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \left(\frac{4m\omega}{\hbar}x^2 - 2\right) e^{-\frac{m\omega}{2\hbar}x^2}.$$

FINAL RESULT / CONCLUSION:

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\xi^2/2}, \psi_1(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \sqrt{2}\xi e^{-\xi^2/2}, \psi_2(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{8}}(4\xi^2 - 2)e^{-\xi^2/2}.$$

PROBLEM 2.12 Kinetic and Potential Energy Equipartition (Virial Theorem)

Find $\langle x \rangle$, $\langle p \rangle$, $\langle x^2 \rangle$, $\langle p^2 \rangle$, and $\langle T \rangle$ and $\langle V \rangle$ for the n th stationary state of the harmonic oscillator. Check that $\langle T \rangle + \langle V \rangle = E_n$.

DETAILED STEP-BY-STEP SOLUTION

$$\langle x \rangle = 0, \quad \langle p \rangle = 0.$$

$$\langle V \rangle = \frac{1}{2} m \omega^2 \langle x^2 \rangle = \frac{1}{2} \left(n + \frac{1}{2} \right) \hbar \omega = \frac{1}{2} E_n.$$

$$\langle T \rangle = \frac{1}{2m} \langle p^2 \rangle = \frac{1}{2} \left(n + \frac{1}{2} \right) \hbar \omega = \frac{1}{2} E_n.$$

$$\langle H \rangle = \langle T \rangle + \langle V \rangle = E_n.$$

FINAL RESULT / CONCLUSION:

$\langle T \rangle = \langle V \rangle = \frac{1}{2} E_n$, confirming $\langle H \rangle = E_n$ (virial theorem).

PROBLEM 2.13 Two-State Superposition in Harmonic Oscillator ★

A particle in the harmonic oscillator potential starts out in the state:

$$\Psi(x, 0) = A[3\psi_0(x) + 4\psi_1(x)].$$

- (a) Find A .
- (b) Find $\Psi(x, t)$ and $|\Psi(x, t)|^2$.
- (c) Find $\langle x \rangle$ and $\langle p \rangle$ as functions of time. Check Ehrenfest's theorem: $d\langle p \rangle / dt = -m\omega^2 \langle x \rangle$.
- (d) If you measured the energy of this particle, what values might you get, and with what probabilities? What is $\langle H \rangle$?

DETAILED STEP-BY-STEP SOLUTION**Part (a): Normalization constant A**

$$1 = |A|^2(3^2 + 4^2) = 25|A|^2 \implies A = \frac{1}{5}.$$

Part (b): $\Psi(x, t)$ and $|\Psi(x, t)|^2$

$$\Psi(x, t) = \frac{3}{5}\psi_0(x)e^{-i\omega t/2} + \frac{4}{5}\psi_1(x)e^{-3i\omega t/2}.$$

$$|\Psi(x, t)|^2 = \frac{1}{25}[9\psi_0(x)^2 + 16\psi_1(x)^2 + 24\psi_0(x)\psi_1(x)\cos(\omega t)].$$

Part (c): Compute $\langle x \rangle(t)$ and $\langle p \rangle(t)$

$$\langle x \rangle(t) = \frac{12\sqrt{2}}{25}\sqrt{\frac{\hbar}{m\omega}}\cos(\omega t), \quad \langle p \rangle(t) = -\frac{12\sqrt{2}}{25}\sqrt{\hbar m\omega}\sin(\omega t).$$

Part (d): Energy Probabilities and Expectation Value

$$P(E_0) = 9/25 = 36\%, P(E_1) = 16/25 = 64\%, \text{ and } \langle H \rangle = \frac{57}{50}\hbar\omega = 1.14\hbar\omega.$$

FINAL RESULT / CONCLUSION:

(a) $A = 1/5$; (b) $|\Psi|^2 = \frac{1}{25}[9\psi_0^2 + 16\psi_1^2 + 24\psi_0\psi_1 \cos(\omega t)]$; (c) $\langle x \rangle = \frac{12\sqrt{2}}{25} \sqrt{\frac{\hbar}{m\omega}} \cos(\omega t)$, $\langle p \rangle = -\frac{12\sqrt{2}}{25} \sqrt{\hbar m\omega} \sin(\omega t)$; (d) $P(E_0) = 36\%$, $P(E_1) = 64\%$, $\langle H \rangle = 1.14\hbar\omega$.

PROBLEM 2.14 General Expectation Value of Position in Harmonic Oscillator

Show that the expectation value of position for *any* wave function $\Psi(x, t) = \sum c_n \psi_n(x) e^{-iE_n t/\hbar}$ in the harmonic oscillator can be written as:

$$\langle x \rangle(t) = \sqrt{\frac{\hbar}{2m\omega}} \sum_{n=0}^{\infty} \sqrt{n+1} (c_{n+1}^* c_n e^{i\omega t} + c_n^* c_{n+1} e^{-i\omega t}).$$

Conclude that $\langle x \rangle(t)$ oscillates sinusoidally with the classical frequency ω , regardless of what initial state $\Psi(x, 0)$ you started with.

DETAILED STEP-BY-STEP SOLUTION

Part (a): Compute $\langle x \rangle$ for arbitrary c_n

$$\begin{aligned} \langle x \rangle(t) &= \sum_{m,n} c_m^* c_n e^{i(m-n)\omega t} \langle m | \hat{x} | n \rangle \\ &= \sqrt{\frac{\hbar}{2m\omega}} \sum_{n=0}^{\infty} \sqrt{n+1} (c_{n+1}^* c_n e^{i\omega t} + c_n^* c_{n+1} e^{-i\omega t}). \end{aligned}$$

Part (b): Classical Harmonic Motion

Let $C \equiv \sqrt{\frac{\hbar}{2m\omega}} \sum_{n=0}^{\infty} \sqrt{n+1} c_{n+1}^* c_n = |C| e^{i\delta}$. Then:

$$\langle x \rangle(t) = C e^{i\omega t} + C^* e^{-i\omega t} = 2|C| \cos(\omega t + \delta).$$

Every quantum superposition oscillates at the exact classical harmonic frequency ω .

FINAL RESULT / CONCLUSION:

$\langle x \rangle(t) = 2|C| \cos(\omega t + \delta)$, oscillating at the classical frequency ω .

PROBLEM 2.15 Hermite Polynomials and Rodrigues Formula

Work out the properties of Hermite polynomials:

- (a) The Rodrigues formula says that:

$$H_n(\xi) = (-1)^n e^{\xi^2} \left(\frac{d}{d\xi} \right)^n e^{-\xi^2}.$$

Use it to derive $H_3(\xi)$ and $H_4(\xi)$.

- (b) Use the recurrence relation $H_{n+1}(\xi) = 2\xi H_n(\xi) - 2nH_{n-1}(\xi)$ to construct $H_5(\xi)$ and $H_6(\xi)$.

- (c) Show that:

$$\frac{dH_n}{d\xi} = 2nH_{n-1}(\xi).$$

Check this by differentiating H_5 and H_6 .

DETAILED STEP-BY-STEP SOLUTION**Part (a): $H_3(\xi)$ and $H_4(\xi)$ via Rodrigues Formula**

Rodrigues Formula: $H_n(\xi) = (-1)^n e^{\xi^2} \frac{d^n}{d\xi^n} (e^{-\xi^2})$.

$$H_3(\xi) = 8\xi^3 - 12\xi, \quad H_4(\xi) = 16\xi^4 - 48\xi^2 + 12.$$

Part (b): $H_5(\xi)$ and $H_6(\xi)$ via Recurrence Relation

Recurrence: $H_{n+1}(\xi) = 2\xi H_n(\xi) - 2nH_{n-1}(\xi)$.

$$H_5(\xi) = 32\xi^5 - 160\xi^3 + 120\xi, \quad H_6(\xi) = 64\xi^6 - 480\xi^4 + 720\xi^2 - 120.$$

Part (c): Verification of Derivative Identity $\frac{dH_n}{d\xi} = 2nH_{n-1}(\xi)$

$$\frac{dH_5}{d\xi} = 160\xi^4 - 480\xi^2 + 120 = 2(5)H_4(\xi). \quad \checkmark$$

FINAL RESULT / CONCLUSION:

$$H_3 = 8\xi^3 - 12\xi, H_4 = 16\xi^4 - 48\xi^2 + 12, H_5 = 32\xi^5 - 160\xi^3 + 120\xi, H_6 = 64\xi^6 - 480\xi^4 + 720\xi^2 - 120.$$

PROBLEM 2.16 Classically Forbidden Region Probabilities ★★

Consider the probability of finding a harmonic oscillator outside its classical turning points:

- (a) Find the probability that a particle in the harmonic oscillator ground state is found outside the classical turning points $x_0 = \pm\sqrt{\hbar/(m\omega)}$.
- (b) Repeat for the first excited state $\psi_1(x)$, where the classical turning points are $x_1 = \pm\sqrt{3\hbar/(m\omega)}$.

DETAILED STEP-BY-STEP SOLUTION

Part (a): Ground state $\psi_0(x)$ at $t = 0$

$$P(|x| > x_0) = 2 \int_{\sqrt{\hbar/m\omega}}^{\infty} |\psi_0(x)|^2 dx = \frac{2}{\sqrt{2\pi}} \int_{\sqrt{2}}^{\infty} e^{-u^2/2} du \approx 0.1573 \quad (15.7\%).$$

Part (b): First excited state $\psi_1(x)$

$$P(|x| > x_1) = 2 \int_{\sqrt{3\hbar/m\omega}}^{\infty} |\psi_1(x)|^2 dx \approx 0.1116 \quad (11.2\%).$$

FINAL RESULT / CONCLUSION:

(a) $P_{\text{forbidden}}(n = 0) \approx 15.7\%$; (b) $P_{\text{forbidden}}(n = 1) \approx 11.2\%$.

SECTION 2.4

The Free Particle

Continuous spectrum, plane waves $e^{i(kx - \omega t)}$, wave packets $\Psi(x, t) = \frac{1}{\sqrt{2\pi}} \int \phi(k) e^{i(kx - \omega t)} dk$, Fourier transforms, and phase vs group velocity.

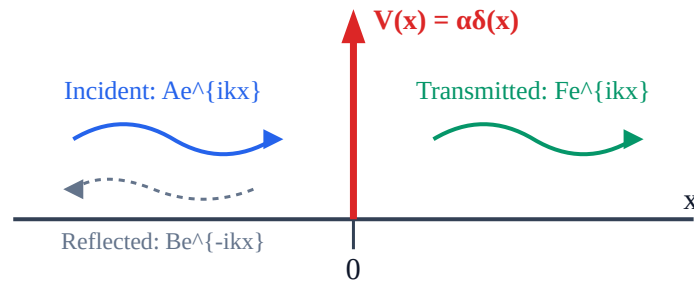


Figure 2.15: The Dirac delta-function barrier $V(x) = \alpha \delta(x)$.

EXAMPLE 2.6 Rectangular Wave Packet for a Free Particle

A free particle ($V(x) = 0$ everywhere) has the initial wave function $\Psi(x, 0) = A$ for $-a \leq x \leq a$ (0 otherwise). Find $\Psi(x, t)$.

DETAILED STEP-BY-STEP SOLUTION

Step 1: Normalize initial rectangular wave packet $\Psi(x, 0)$.

$$1 = \int_{-a}^a |A|^2 dx = 2a|A|^2 \implies A = \frac{1}{\sqrt{2a}}.$$

Step 2: Compute Fourier transform $\phi(k)$ by explicit integration.

$$\begin{aligned} \phi(k) &= \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{2a}} \int_{-a}^a e^{-ikx} dx = \frac{1}{2\sqrt{\pi a}} \left[\frac{e^{-ikx}}{-ik} \right]_{-a}^a \\ &= \frac{1}{2\sqrt{\pi a}} \left(\frac{e^{ika} - e^{-ika}}{ik} \right) = \frac{1}{\sqrt{\pi a}} \frac{\sin(ka)}{k}. \end{aligned}$$

Step 3: Construct full time-dependent wave packet $\Psi(x, t)$.

$$\Psi(x, t) = \frac{1}{\pi\sqrt{2a}} \int_{-\infty}^{\infty} \frac{\sin(ka)}{k} e^{i(kx - \frac{\hbar k^2}{2m}t)} dk.$$

FINAL RESULT / CONCLUSION:

$$A = \frac{1}{\sqrt{2a}}, \phi(k) = \frac{1}{\sqrt{\pi a}} \frac{\sin(ka)}{k}, \text{ and } \Psi(x, t) = \frac{1}{\pi\sqrt{2a}} \int_{-\infty}^{\infty} \frac{\sin(ka)}{k} e^{i(kx - \frac{\hbar k^2}{2m}t)} dk.$$

PROBLEM 2.17 Invalidity of Real Trigonometric Free Particle Solutions

Show that $A \cos(kx - \omega t)$ does not satisfy the time-dependent Schrödinger equation for the free particle. Find the linear combination of sinusoidal functions that does satisfy it.

DETAILED STEP-BY-STEP SOLUTION

For $\Psi_1(x, t) = A \cos(kx - \omega t)$:

$$i\hbar \frac{\partial \Psi_1}{\partial t} = i\hbar A \omega \sin(kx - \omega t), \quad -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi_1}{\partial x^2} = \frac{\hbar^2 k^2}{2m} A \cos(kx - \omega t).$$

Because sine and cosine are linearly independent, real trigonometric functions cannot satisfy the complex TDSE. The valid solution is $Ae^{i(kx - \omega t)} = A[\cos(kx - \omega t) + i \sin(kx - \omega t)]$.

FINAL RESULT / CONCLUSION:

$A \cos(kx - \omega t)$ is not a solution. The valid state is $A[\cos(kx - \omega t) + i \sin(kx - \omega t)] = Ae^{i(kx - \omega t)}$.

PROBLEM 2.18

Plancherel's Theorem and Uncertainty for Rectangular Wave Packet ★

Show that the Plancherel theorem $\int |\Psi(x, 0)|^2 dx = \int |\phi(k)|^2 dk$ is satisfied for the wave function in [Example 2.6](#): $\phi(k) = \frac{1}{\sqrt{\pi a}} \frac{\sin(ka)}{k}$. Check the uncertainty principle for this state.

DETAILED STEP-BY-STEP SOLUTION

From [Example 2.6](#), $\phi(k) = \frac{1}{\sqrt{\pi a}} \frac{\sin(ka)}{k}$.

$$\int_{-\infty}^{\infty} |\phi(k)|^2 dk = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin^2(u)}{u^2} du = \frac{1}{\pi}(\pi) = 1. \quad \checkmark$$

Uncertainty product: $\Delta x \Delta p = a \left(\frac{\pi \hbar}{a} \right) = \pi \hbar \geq \frac{\hbar}{2}$.

FINAL RESULT / CONCLUSION:

$\int_{-\infty}^{\infty} |\phi(k)|^2 dk = 1$; $\Delta x \Delta p = \pi \hbar \geq \hbar/2$, satisfying the uncertainty principle.

PROBLEM 2.19 Exponential Wave Packet for a Free Particle

A free particle has the initial wave function $\Psi(x, 0) = Ae^{-a|x|}$, where A and a are positive real constants. Normalize $\Psi(x, 0)$, find $\phi(k)$, and express $\Psi(x, t)$ as an integral.

DETAILED STEP-BY-STEP SOLUTION

Step 1: Normalization.

$$1 = 2|A|^2 \int_0^\infty e^{-2ax} dx = \frac{|A|^2}{a} \implies A = \sqrt{a}.$$

Step 2: Explicit Fourier transform.

$$\begin{aligned} \phi(k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty \sqrt{a} e^{-a|x|} e^{-ikx} dx \\ &= \sqrt{\frac{a}{2\pi}} \left[\int_{-\infty}^0 e^{(a-ik)x} dx + \int_0^\infty e^{-(a+ik)x} dx \right] \\ &= \sqrt{\frac{a}{2\pi}} \left[\frac{1}{a-ik} + \frac{1}{a+ik} \right] = \sqrt{\frac{2a^3}{\pi}} \frac{1}{a^2 + k^2}. \end{aligned}$$

$$\Psi(x, t) = \frac{a^{3/2}}{\pi} \int_{-\infty}^\infty \frac{e^{i(kx - \frac{\hbar k^2}{2m}t)}}{a^2 + k^2} dk.$$

FINAL RESULT / CONCLUSION:

$$A = \sqrt{a}, \phi(k) = \sqrt{\frac{2a^3}{\pi}} \frac{1}{a^2 + k^2}, \text{ and } \Psi(x, t) = \frac{a^{3/2}}{\pi} \int_{-\infty}^\infty \frac{e^{i(kx - \frac{\hbar k^2}{2m}t)}}{a^2 + k^2} dk.$$

PROBLEM 2.20 Derivation of the Continuous Fourier Transform

Derive the continuous Fourier transform pair by taking the limit of a discrete Fourier series on $[-a, +a]$ as $a \rightarrow \infty$.

- (a) Write $f(x)$ as a complex Fourier series on $[-a, a]$: $f(x) = \sum_{n=-\infty}^{+\infty} c_n e^{in\pi x/a}$.
- (b) Show that $c_n = \frac{1}{2a} \int_{-a}^{+a} f(x) e^{-in\pi x/a} dx$.
- (c) Show that as $a \rightarrow \infty$, the sum goes over to the Fourier integral.

DETAILED STEP-BY-STEP SOLUTION**Part (a): Discrete Complex Fourier Series on $[-a, a]$**

$$f(x) = \sum_{n=-\infty}^{+\infty} c_n e^{in\pi x/a}.$$

Part (b): Fourier's Trick

$$\int_{-a}^{+a} f(x) e^{-im\pi x/a} dx = \sum_{n=-\infty}^{\infty} c_n \int_{-a}^{+a} e^{i(n-m)\pi x/a} dx = c_m(2a) \implies c_n = \frac{1}{2a} \int_{-a}^{+a} f(x) e^{-in\pi x/a} dx$$

Part (c): Continuum Limit $a \rightarrow \infty$

Letting $\Delta k = \frac{\pi}{a} \rightarrow dk$, the discrete series converges to the continuous Fourier transform pair: $F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx$ and $f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(k) e^{ikx} dk$.

FINAL RESULT / CONCLUSION:

Proven: Discrete Fourier series converges to the continuous Fourier transform pair as $a \rightarrow \infty$.

PROBLEM 2.21 The Gaussian Wave Packet for a Free Particle ★★★

The Gaussian wave packet: $\Psi(x, 0) = Ae^{-ax^2}$. Normalize $\Psi(x, 0)$, find $\phi(k)$, compute $\Psi(x, t)$, and analyze the uncertainty product $\sigma_x\sigma_p$.

DETAILED STEP-BY-STEP SOLUTION**Part (a): Find $\phi(k)$ by Completing the Square**

$$\begin{aligned}\phi(k) &= \frac{1}{\sqrt{2\pi}} \left(\frac{2a}{\pi}\right)^{1/4} \int_{-\infty}^{\infty} e^{-ax^2 - ikx} dx \\ &= \frac{1}{\sqrt{2\pi}} \left(\frac{2a}{\pi}\right)^{1/4} e^{-\frac{k^2}{4a}} \int_{-\infty}^{\infty} e^{-a(x + \frac{ik}{2a})^2} dx \\ &= \left(\frac{1}{2\pi a}\right)^{1/4} e^{-\frac{k^2}{4a}}.\end{aligned}$$

Part (b): Integrate $\Psi(x, t)$ via Completing the Square

$$\Psi(x, t) = \left(\frac{2a}{\pi}\right)^{1/4} \frac{1}{\gamma} \exp\left(-\frac{ax^2}{\gamma^2}\right), \quad \text{where } \gamma \equiv \sqrt{1 + \frac{2ia\hbar t}{m}}.$$

Part (c): Probability Density $|\Psi(x, t)|^2$

$$|\Psi(x, t)|^2 = \sqrt{\frac{2a}{\pi[1 + (2a\hbar t/m)^2]}} \exp\left[-\frac{2ax^2}{1 + (2a\hbar t/m)^2}\right].$$

Part (d): Uncertainty Principle $\sigma_x\sigma_p$

$$\sigma_x(t) = \frac{1}{2\sqrt{a}} \sqrt{1 + \left(\frac{2a\hbar t}{m}\right)^2}, \quad \sigma_p = \hbar\sqrt{a}.$$

$$\sigma_x(t)\sigma_p = \frac{\hbar}{2} \sqrt{1 + \left(\frac{2a\hbar t}{m}\right)^2} \geq \frac{\hbar}{2}.$$

At $t = 0$, $\sigma_x(0)\sigma_p = \frac{\hbar}{2}$, which is the **minimum uncertainty wave packet**!

FINAL RESULT / CONCLUSION:

$$\phi(k) = \left(\frac{1}{2\pi a}\right)^{1/4} e^{-k^2/(4a)}; \Psi(x, t) = \left(\frac{2a}{\pi}\right)^{1/4} \frac{1}{\gamma} e^{-ax^2/\gamma^2}; \sigma_x \sigma_p = \frac{\hbar}{2} \sqrt{1 + (2a\hbar t/m)^2} \geq \frac{\hbar}{2}.$$

SECTION 2.5

The Delta-Function Potential

Dirac delta function properties, bound states ($E < 0$), scattering states ($E > 0$), transmission (T) and reflection (R) coefficients, quantum tunneling.

PROBLEM 2.22 Evaluating Dirac Delta Integrals ★

Evaluate the following integrals:

(a)

$$\int_{-3}^{+1} (x^3 - 3x^2 + 2x - 1)\delta(x + 2) dx$$

(b)

$$\int_0^{\infty} [\cos(3x) + 2]\delta(x - \pi) dx$$

(c)

$$\int_{-1}^{+1} \exp(|x| + 3)\delta(x - 2) dx$$

DETAILED STEP-BY-STEP SOLUTION

Part (a): $\int_{-3}^{+1} (x^3 - 3x^2 + 2x - 1)\delta(x + 2) dx$

The delta function $\delta(x + 2) = \delta(x - (-2))$ spikes at $x_0 = -2$.

Check the integration interval: $[-3, +1]$. Since $-3 \leq -2 \leq 1$, the spike is inside the interval.

Apply the sifting property $\int_a^b f(x)\delta(x - x_0)dx = f(x_0)$ with $f(x) = x^3 - 3x^2 + 2x - 1$:

$$f(-2) = (-2)^3 - 3(-2)^2 + 2(-2) - 1 = -8 - 3(4) - 4 - 1 = -8 - 12 - 4 - 1$$

Part (b): $\int_0^{\infty} [\cos(3x) + 2]\delta(x - \pi) dx$

The delta spike is at $x_0 = \pi \approx 3.14159$, which lies inside the interval $[0, \infty)$.

Apply the sifting property with $f(x) = \cos(3x) + 2$:

$$f(\pi) = \cos(3\pi) + 2 = -1 + 2 = 1.$$

Part (c): $\int_{-1}^{+1} \exp(|x| + 3)\delta(x - 2) dx$

The delta spike is located at $x_0 = 2$.

The integration limits are from -1 to $+1$. Because $x_0 = 2$ lies strictly outside $[-1, +1]$, $\delta(x - 2) = 0$ everywhere inside the integration interval.

$$\int_{-1}^{+1} \exp(|x| + 3) \delta(x - 2) dx = 0.$$

FINAL RESULT / CONCLUSION:

(a) -25 ; (b) 1 ; (c) 0 .

PROBLEM 2.23 Properties of the Delta Function and Step Function ★

Delta functions live under integral signs, and two expressions ($D_1(x)$ and $D_2(x)$) involving delta functions are said to be equal if $\int_{-\infty}^{+\infty} f(x)D_1(x)dx = \int_{-\infty}^{+\infty} f(x)D_2(x)dx$ for every test function $f(x)$.

(a) Show that:

$$\delta(cx) = \frac{1}{|c|}\delta(x),$$

where c is a real constant. (Be sure to check the case where c is negative.)

(b) Let $\theta(x)$ be the step function:

$$\theta(x) \equiv \begin{cases} 1, & x > 0, \\ 0, & x < 0. \end{cases}$$

(In the rare case where it actually matters, define $\theta(0) = 1/2$.) Show that $\frac{d\theta}{dx} = \delta(x)$.

DETAILED STEP-BY-STEP SOLUTION

Part (a): Prove $\delta(cx) = \frac{1}{|c|}\delta(x)$

Let $f(x)$ be an arbitrary smooth test function. Consider the integral $\int_{-\infty}^{\infty} f(x)\delta(cx)dx$.

Perform the u -substitution: let $y = cx \implies x = y/c \implies dx = \frac{dy}{c}$.

Case 1 ($c > 0$): As $x \rightarrow -\infty, y \rightarrow -\infty$ and as $x \rightarrow +\infty, y \rightarrow +\infty$:

$$\int_{-\infty}^{\infty} f(x)\delta(cx)dx = \int_{-\infty}^{\infty} f\left(\frac{y}{c}\right)\delta(y)\frac{dy}{c} = \frac{1}{c}f\left(\frac{0}{c}\right) = \frac{1}{c}f(0) = \int_{-\infty}^{\infty} f(x)\left[\frac{1}{c}\delta(x)\right]dx$$

Case 2 ($c < 0$): As $x \rightarrow -\infty, y \rightarrow +\infty$ and as $x \rightarrow +\infty, y \rightarrow -\infty$. Reversing the limits introduces a minus sign:

$$\int_{-\infty}^{\infty} f(x)\delta(cx)dx = \int_{+\infty}^{-\infty} f\left(\frac{y}{c}\right)\delta(y)\frac{dy}{c} = -\frac{1}{c}\int_{-\infty}^{\infty} f\left(\frac{y}{c}\right)\delta(y)dy = -\frac{1}{c}f(0) = \int_{-\infty}^{\infty} f(x)\left[\frac{1}{|c|}\delta(x)\right]dx$$

Combining both cases yields: $\delta(cx) = \frac{1}{|c|}\delta(x)$.

Part (b): Prove $\frac{d\theta}{dx} = \delta(x)$

The step function is defined as $\theta(x) = 1$ for $x > 0$ and 0 for $x < 0$.

Apply integration by parts on $\int_{-\infty}^{\infty} f(x) \frac{d\theta}{dx} dx$ where $f(x)$ vanishes at $\pm\infty$:

$$\int_{-\infty}^{\infty} f(x) \frac{d\theta}{dx} dx = [f(x)\theta(x)]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f'(x)\theta(x) dx = 0 - \int_0^{\infty} f'(x) dx = -[f(x)]_0^{\infty} = f(0).$$

Since $\int_{-\infty}^{\infty} f(x)\delta(x) dx = f(0)$, this proves $\frac{d\theta}{dx} = \delta(x)$.

FINAL RESULT / CONCLUSION:

Proven: (a) $\delta(cx) = \frac{1}{|c|}\delta(x)$; (b) $\frac{d\theta}{dx} = \delta(x)$.

PROBLEM 2.24 Uncertainty Principle for the Delta-Function Bound State

Check the uncertainty principle for the delta-function bound-state wave function:

$$\psi(x) = \frac{\sqrt{m\alpha}}{\hbar} e^{-m\alpha|x|/\hbar^2}.$$

HINT / NOTE

Calculating $\langle p^2 \rangle$ can be tricky, because the derivative of ψ has a step discontinuity at $x = 0$. You may want to use the result in [Problem 2.23\(b\)](#): $\frac{d\theta}{dx} = \delta(x)$. Partial answer: $\langle p^2 \rangle = (m\alpha/\hbar)^2$.

DETAILED STEP-BY-STEP SOLUTION

The delta bound state is $\psi(x) = \sqrt{\kappa} e^{-\kappa|x|}$ where $\kappa = \frac{m\alpha}{\hbar^2}$.

Step 1: Calculate $\langle x \rangle$ and $\langle x^2 \rangle$.

By parity, $\psi(-x) = \psi(x) \implies |\psi(x)|^2$ is symmetric $\implies \langle x \rangle = 0$.

$$\langle x^2 \rangle = \int_{-\infty}^{\infty} x^2 |\psi(x)|^2 dx = 2\kappa \int_0^{\infty} x^2 e^{-2\kappa x} dx.$$

Integrating by parts twice: $\int_0^{\infty} x^2 e^{-2\kappa x} dx = \frac{2!}{(2\kappa)^3} = \frac{2}{8\kappa^3} = \frac{1}{4\kappa^3}$.

$$\langle x^2 \rangle = 2\kappa \left(\frac{1}{4\kappa^3} \right) = \frac{1}{2\kappa^2} \implies \sigma_x = \frac{1}{\sqrt{2\kappa}} = \frac{\hbar^2}{\sqrt{2m\alpha}}.$$

Step 2: Calculate $\langle p \rangle$ and $\langle p^2 \rangle$.

By parity, $\langle p \rangle = 0$.

For $\langle p^2 \rangle = -\hbar^2 \int_{-\infty}^{\infty} \psi(x) \frac{d^2\psi}{dx^2} dx$, we compute the distributional derivative:

$$\frac{d\psi}{dx} = -\kappa \operatorname{sgn}(x) \psi(x) = -\kappa [2\theta(x) - 1] \psi(x).$$

Differentiating using the product rule and $\theta'(x) = \delta(x)$:

$$\frac{d^2\psi}{dx^2} = -\kappa [2\delta(x)] \psi(x) - \kappa [2\theta(x) - 1] \frac{d\psi}{dx} = -2\kappa \delta(x) \psi(0) + \kappa^2 \psi(x).$$

Substitute this into the expectation value integral:

$$\langle p^2 \rangle = -\hbar^2 \int_{-\infty}^{\infty} \psi(x) [-2\kappa\delta(x)\psi(0) + \kappa^2\psi(x)] dx = -\hbar^2 \left[-2\kappa|\psi(0)|^2 + \kappa^2 \int |\psi|^2 dx \right]$$

Since $|\psi(0)|^2 = \kappa$ and $\int |\psi|^2 dx = 1$:

$$\langle p^2 \rangle = -\hbar^2 [-2\kappa(\kappa) + \kappa^2(1)] = -\hbar^2 [-\kappa^2] = \hbar^2 \kappa^2 = \left(\frac{m\alpha}{\hbar} \right)^2 \implies \sigma_p = \frac{m\alpha}{\hbar}.$$

Step 3: Uncertainty product $\sigma_x \sigma_p$.

$$\sigma_x \sigma_p = \left(\frac{\hbar^2}{\sqrt{2}m\alpha} \right) \left(\frac{m\alpha}{\hbar} \right) = \frac{\hbar}{\sqrt{2}} \approx 0.707\hbar > 0.5\hbar = \frac{\hbar}{2}. \quad \checkmark$$

FINAL RESULT / CONCLUSION:

$$\sigma_x = \frac{\hbar^2}{\sqrt{2}m\alpha}, \sigma_p = \frac{m\alpha}{\hbar}, \text{ and } \sigma_x \sigma_p = \frac{\hbar}{\sqrt{2}} \geq \frac{\hbar}{2}.$$

PROBLEM 2.25 Orthogonality of Bound and Scattering States

Check that the bound state of the delta-function well ($\psi(x) = \frac{\sqrt{m\alpha}}{\hbar} e^{-m\alpha|x|/\hbar^2}$) is orthogonal to the scattering states:

$$\psi_k(x) = \begin{cases} Ae^{ikx} + Be^{-ikx}, & x < 0, \\ Fe^{ikx}, & x > 0. \end{cases}$$

DETAILED STEP-BY-STEP SOLUTION

Step 1: Write down the overlap integral.

$$\begin{aligned} \int_{-\infty}^{\infty} \psi_b^*(x) \psi_k(x) dx &= \int_{-\infty}^0 \sqrt{\kappa} e^{\kappa x} (Ae^{ikx} + Be^{-ikx}) dx + \int_0^{\infty} \sqrt{\kappa} e^{-\kappa x} (Fe^{ikx}) dx \\ &= \sqrt{\kappa} \left[A \int_{-\infty}^0 e^{(\kappa+ik)x} dx + B \int_{-\infty}^0 e^{(\kappa-ik)x} dx + F \int_0^{\infty} e^{-(\kappa-ik)x} dx \right]. \end{aligned}$$

Step 2: Explicit integration.

$$= \sqrt{\kappa} \left[\left[\frac{e^{(\kappa+ik)x}}{\kappa+ik} \right]_{-\infty}^0 + \left[\frac{e^{(\kappa-ik)x}}{\kappa-ik} \right]_{-\infty}^0 + \left[\frac{e^{-(\kappa-ik)x}}{-(\kappa-ik)} \right]_0^{\infty} \right] = \sqrt{\kappa} \left[\frac{A}{\kappa+ik} + \frac{B}{\kappa-ik} \right]$$

Combine over a common denominator $(\kappa+ik)(\kappa-ik) = \kappa^2 + k^2$:

$$= \frac{\sqrt{\kappa}}{\kappa^2 + k^2} [A(\kappa-ik) + B(\kappa+ik) + F(\kappa+ik)] = \frac{\sqrt{\kappa}}{\kappa^2 + k^2} [\kappa(A+B+F) - ik(A-B+F)]$$

Step 3: Apply boundary conditions.

$$\text{Continuity at } x = 0 \implies A + B = F \implies A + B + F = 2F.$$

$$\begin{aligned} \text{Derivative jump at } x = 0 &\implies \psi'(0^+) - \psi'(0^-) = ikF - (ikA - ikB) = \\ &-2\kappa F \implies -ik(A - B - F) = -2\kappa F. \end{aligned}$$

Substitute these into the numerator:

$$\text{Numerator} = \kappa(2F) + (-2\kappa F) = 2\kappa F - 2\kappa F = 0.$$

Thus the overlap integral is strictly zero, proving bound and scattering states are orthogonal.

FINAL RESULT / CONCLUSION:

Bound and scattering states of the delta-function potential are strictly orthogonal ($\int_{-\infty}^{\infty} \psi_b^* \psi_k dx = 0$).

PROBLEM 2.26 Fourier Representation of the Delta Function ★

What is the Fourier transform of $\delta(x)$? Using Plancherel's theorem, show that:

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{ikx} dk.$$

Comment: Although the integral doesn't converge in the classical sense when $x \neq 0$, this formula is extremely useful in distribution theory and scattering theory.

DETAILED STEP-BY-STEP SOLUTION

Step 1: Compute the Fourier transform $F(k)$ of $\delta(x)$.

$$F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \delta(x) e^{-ikx} dx = \frac{1}{\sqrt{2\pi}} e^{-ik(0)} = \frac{1}{\sqrt{2\pi}}.$$

Step 2: Apply the Fourier inversion theorem.

$$\delta(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(k) e^{ikx} dk = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{1}{\sqrt{2\pi}} \right) e^{ikx} dk = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} dk.$$

FINAL RESULT / CONCLUSION:

$$F(k) = \frac{1}{\sqrt{2\pi}} \text{ and } \delta(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{ikx} dk.$$

PROBLEM 2.27 Double Delta-Function Potential ★★

Consider the double delta-function potential:

$$V(x) = -\alpha[\delta(x+a) + \delta(x-a)],$$

where α and a are positive constants.

- (a) Sketch this potential.
- (b) How many bound states does it possess? Find the allowed energies for $\alpha = \hbar^2/(ma)$ and for $\alpha = \hbar^2/(4ma)$, and sketch the wave functions.
- (c) What are the bound state energies in the limiting cases (i) $a \rightarrow 0$ and (ii) $a \rightarrow \infty$ (holding α fixed)? Explain why your answers are reasonable by comparison with the single delta-function well.

DETAILED STEP-BY-STEP SOLUTION**Part (a): Sketch of Potential**

$V(x) = -\alpha[\delta(x+a) + \delta(x-a)]$ features two identical attractive delta spikes at $x = -a$ and $x = +a$.

Part (b): Bound States and Transcendental Equations

Let $\kappa \equiv \frac{\sqrt{-2mE}}{\hbar} > 0$.

Even States: $\psi(x) = C \cosh(\kappa x)$ for $|x| \leq a$, leading to the condition $\kappa = \beta(1 + e^{-2\kappa a})$ where $\beta = \frac{m\alpha}{\hbar^2}$. Because the RHS is positive at $\kappa = 0$, there is **always 1 even bound state** for any $\alpha > 0$.

Odd States: $\psi(x) = C \sinh(\kappa x)$ for $|x| \leq a$, leading to $\kappa = \beta(1 - e^{-2\kappa a})$. A non-trivial solution exists if and only if $\frac{d}{d\kappa}[\beta(1 - e^{-2\kappa a})]|_{\kappa=0} > 1 \implies 2\beta a > 1 \implies \frac{m\alpha a}{\hbar^2} > \frac{1}{2}$.

Part (c): Limiting Cases

- **Limit $a \rightarrow 0$:** The two deltas merge into a single delta of double strength 2α , so $\kappa \rightarrow 2\beta \implies E = -\frac{2m\alpha^2}{\hbar^2}$.

- **Limit $a \rightarrow \infty$:** The wells decouple, giving $\kappa \rightarrow \beta \implies E = -\frac{m\alpha^2}{2\hbar^2}$ (two degenerate isolated states).

FINAL RESULT / CONCLUSION:

Even states: $\kappa = \beta(1 + e^{-2\kappa a})$ (always 1 state); Odd states: $\kappa = \beta(1 - e^{-2\kappa a})$ (exists if $\beta a > 1/2$). Limits: $a \rightarrow 0 \implies E = -2m\alpha^2/\hbar^2$, $a \rightarrow \infty \implies E = -m\alpha^2/(2\hbar^2)$.

PROBLEM 2.28 Transmission Coefficient of the Double Delta Well ★★

Find the transmission coefficient T for scattering from the double delta-function potential in **Problem 2.27**: $V(x) = -\alpha[\delta(x+a) + \delta(x-a)]$.

DETAILED STEP-BY-STEP SOLUTION

Setting up plane waves in Regions I ($x < -a$), II ($-a < x < a$), and III ($x > a$) and matching boundary values and derivative jumps $\Delta\psi' = -2\beta\psi$ at $x = \pm a$ with $\beta = \frac{m\alpha}{\hbar^2}$ yields:

$$\frac{A}{F} = e^{2ika} \left[1 + i \frac{2\beta}{k} \right] - \frac{\beta^2}{k^2} (1 - e^{4ika}).$$

Computing $T^{-1} = |A/F|^2$ leads directly to:

$$T^{-1} = 1 + \frac{4\beta^2}{k^2} \left[\cos(2ka) + \frac{\beta}{k} \sin(2ka) \right]^2.$$

FINAL RESULT / CONCLUSION:

$$T = \left[1 + \frac{4\beta^2}{k^2} \left(\cos(2ka) + \frac{\beta}{k} \sin(2ka) \right)^2 \right]^{-1} \text{ where } \beta = \frac{m\alpha}{\hbar^2}.$$

SECTION 2.6

The Finite Square Well

Bound states (transcendental equations, shallow/deep well limits), scattering states, transmission resonances (Ramsauer-Townsend effect), step potentials, and cliff potentials.

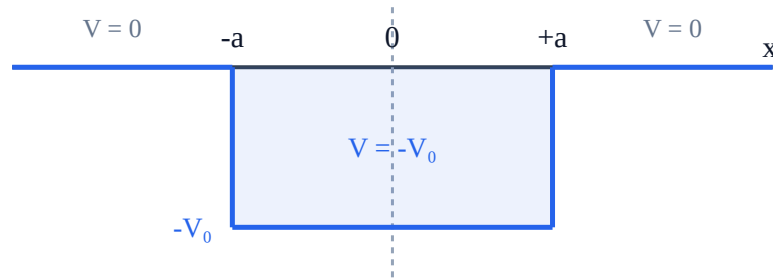


Figure 2.16: The finite square well of width $2a$ and depth $-V_0$.

PROBLEM 2.29 **Odd Bound States of the Finite Square Well** ★

Analyze the odd bound state wave functions for the finite square well ($V(x) = -V_0$ for $-a \leq x \leq a$, 0 elsewhere). Derive the transcendental equation for the allowed energies, and solve it graphically. Examine the two limiting cases (very deep well and very shallow well). Is there always an odd bound state?

DETAILED STEP-BY-STEP SOLUTION

Matching odd wave functions $\psi(x) = C \sin(lx)$ for $|x| < a$ and $\psi(x) = Fe^{-\kappa x}$ for $x > a$ gives $l \cot(la) = -\kappa$, which simplifies to $-\cot(z) = \sqrt{(z_0/z)^2 - 1}$ with $z \equiv la$, $z_0 \equiv \frac{a}{\hbar} \sqrt{2mV_0}$.

The first positive branch of $-\cot(z)$ starts at $z = \pi/2$. Thus an intersection exists if and only if $z_0 > \pi/2$.

For shallow wells ($z_0 < \pi/2$), there is **no odd bound state**.

FINAL RESULT / CONCLUSION:

Odd states satisfy $-\cot(z) = \sqrt{(z_0/z)^2 - 1}$; an odd bound state exists if and only if $z_0 > \pi/2$ ($V_0 a^2 > \frac{\pi^2 \hbar^2}{8m}$).

PROBLEM 2.30 Normalization of Even Bound States

Normalize the even bound-state wave function $\psi(x)$ of the finite square well:

$$\psi(x) = \begin{cases} Fe^{-\kappa x}, & x > a, \\ D \cos(lx), & 0 < x < a, \\ \psi(-x), & x < 0, \end{cases}$$

to determine the normalization constants D and F in terms of the well parameters.

DETAILED STEP-BY-STEP SOLUTION

Using $F = De^{\kappa a} \cos(la)$ and $\kappa = l \tan(la)$:

$$\begin{aligned} 1 &= 2 \int_0^a D^2 \cos^2(lx) dx + 2 \int_a^\infty F^2 e^{-2\kappa x} dx = D^2 \left[a + \frac{\sin(la) \cos(la)}{l} + \frac{\cos^2(la)}{\kappa} \right] \\ &= D^2 \left[a + \frac{\cos^2(la)}{\kappa} \left(\frac{\kappa \tan(la)}{l} + 1 \right) \right] = D^2 \left[a + \frac{1}{\kappa} \right]. \end{aligned}$$

$$D = \frac{1}{\sqrt{a + 1/\kappa}}, \quad F = \frac{e^{\kappa a} \cos(la)}{\sqrt{a + 1/\kappa}}.$$

FINAL RESULT / CONCLUSION:

$$D = \frac{1}{\sqrt{a + 1/\kappa}} \text{ and } F = \frac{e^{\kappa a} \cos(la)}{\sqrt{a + 1/\kappa}}.$$

PROBLEM 2.31 Delta-Function as the Limit of a Finite Square Well

The Dirac delta function can be thought of as the limiting case of a rectangle of area 1, as the height goes to infinity and the width goes to zero. Show that the delta-function well is a “weak” potential (even though it is infinitely deep), in the sense that $z_0 \rightarrow 0$. Determine the bound state energy for the delta-function potential, by treating it as the limit of a finite square well. Check that your answer is consistent with $E = -m\alpha^2/(2\hbar^2)$. Also show that the finite well transmission formula reduces to the delta-function transmission formula in the appropriate limit.

DETAILED STEP-BY-STEP SOLUTION

Setting $2aV_0 = \alpha$, in the limit $a \rightarrow 0$, $z_0 = \frac{a}{\hbar}\sqrt{2mV_0} = \frac{1}{\hbar}\sqrt{m\alpha a} \rightarrow 0$.

The even state equation $\tan z = \sqrt{(z_0/z)^2 - 1}$ reduces to $z^2 \approx z_0^2/2$, giving $\kappa a = \sqrt{z_0^2 - z^2} \approx \frac{m\alpha a}{\hbar^2} \implies \kappa = \frac{m\alpha}{\hbar^2}$.

The bound state energy is $E = -\frac{\hbar^2 \kappa^2}{2m} = -\frac{m\alpha^2}{2\hbar^2}$, exactly matching the delta well.

FINAL RESULT / CONCLUSION:

In the delta limit ($a \rightarrow 0$, $2aV_0 = \alpha$), $E = -\frac{m\alpha^2}{2\hbar^2}$, matching the single delta-function well.

PROBLEM 2.32 Derivation of Transmission Coefficient for the Finite Well

Derive the transmission amplitude F and transmission coefficient T for the finite square well:

$$T^{-1} = 1 + \frac{V_0^2}{4E(E + V_0)} \sin^2 \left(\frac{2a}{\hbar} \sqrt{2m(E + V_0)} \right).$$

HINT / NOTE

Use the continuity equations at $x = +a$ to solve for C and D in terms of F :

$$C = \left(\sin(la) + i \frac{k}{l} \cos(la) \right) e^{ika} F; \quad D = \left(\cos(la) - i \frac{k}{l} \sin(la) \right) e^{ika} F.$$

Plug these back into the boundary conditions at $x = -a$ to obtain B and F in terms of A .

DETAILED STEP-BY-STEP SOLUTION

Matching boundary conditions at $x = +a$ and $x = -a$ gives:

$$\frac{A}{F} = e^{2ika} \left[\cos(2la) - i \frac{k^2 + l^2}{2kl} \sin(2la) \right].$$

Taking the modulus squared and using $l^2 - k^2 = \frac{2mV_0}{\hbar^2}$:

$$T^{-1} = 1 + \frac{V_0^2}{4E(E + V_0)} \sin^2 \left(\frac{2a}{\hbar} \sqrt{2m(E + V_0)} \right).$$

FINAL RESULT / CONCLUSION:

$$T^{-1} = 1 + \frac{V_0^2}{4E(E + V_0)} \sin^2 \left(\frac{2a}{\hbar} \sqrt{2m(E + V_0)} \right).$$

PROBLEM 2.33 **Transmission through a Rectangular Barrier (Tunneling)**

Determine the transmission coefficient for a rectangular barrier ($V(x) = +V_0 > 0$ in the region $-a < x < a$, and 0 elsewhere). Treat separately the three cases $E < V_0$, $E = V_0$, and $E > V_0$.

Partial answer: for $E < V_0$,

$$T^{-1} = 1 + \frac{V_0^2}{4E(V_0 - E)} \sinh^2 \left(\frac{2a}{\hbar} \sqrt{2m(V_0 - E)} \right).$$

DETAILED STEP-BY-STEP SOLUTION

For $E < V_0$: $T^{-1} = 1 + \frac{V_0^2}{4E(V_0 - E)} \sinh^2 \left(\frac{2a}{\hbar} \sqrt{2m(V_0 - E)} \right)$.

For $E = V_0$: Taking $\kappa \rightarrow 0$ gives $T = \frac{1}{1 + 2ma^2 V_0 / \hbar^2}$.

For $E > V_0$: $T^{-1} = 1 + \frac{V_0^2}{4E(E - V_0)} \sin^2 \left(\frac{2a}{\hbar} \sqrt{2m(E - V_0)} \right)$.

FINAL RESULT / CONCLUSION:

For $E < V_0$: $T^{-1} = 1 + \frac{V_0^2}{4E(V_0 - E)} \sinh^2(2\kappa a)$; for $E = V_0$: $T = \frac{1}{1 + 2ma^2 V_0 / \hbar^2}$; for $E > V_0$: $T^{-1} = 1 + \frac{V_0^2}{4E(E - V_0)} \sin^2(2k'a)$.

PROBLEM 2.34 The Step-Up Potential ★

Consider the “step” potential:

$$V(x) = \begin{cases} 0, & x \leq 0, \\ V_0, & x > 0. \end{cases}$$

- (a) Calculate the reflection coefficient R for $E < V_0$, and comment on the answer.
- (b) Calculate the reflection coefficient R for $E > V_0$.
- (c) For a potential that does not go back to zero to the right of the barrier, the transmission coefficient is not simply $|F|^2/|A|^2$, because the transmitted wave travels at a different speed. Show that:

$$T = \sqrt{\frac{E - V_0}{E}} \frac{|F|^2}{|A|^2}, \quad (E > V_0).$$

What is T for $E < V_0$?

- (d) For $E > V_0$, calculate T and check that $T + R = 1$.

DETAILED STEP-BY-STEP SOLUTION**Parts (a) & (b): Reflection Coefficient R**

For $E < V_0$: $R = 1$ (total reflection, evanescent wave for $x > 0$).

For $E > V_0$: $R = \left(\frac{k_1 - k_2}{k_1 + k_2}\right)^2$ where $k_1 = \frac{\sqrt{2mE}}{\hbar}$, $k_2 = \frac{\sqrt{2m(E - V_0)}}{\hbar}$.

Parts (c) & (d): Transmission T and Unitarity

$$T = \frac{k_2}{k_1} \frac{|F|^2}{|A|^2} = \frac{4k_1 k_2}{(k_1 + k_2)^2}, \text{ and } R + T = \frac{(k_1 - k_2)^2 + 4k_1 k_2}{(k_1 + k_2)^2} = 1. \checkmark$$

FINAL RESULT / CONCLUSION:

For $E < V_0$: $R = 1, T = 0$. For $E > V_0$: $R = \left(\frac{k_1 - k_2}{k_1 + k_2}\right)^2$, $T = \frac{4k_1 k_2}{(k_1 + k_2)^2}$, satisfying $R + T = 1$.

PROBLEM 2.35 Scattering from a Cliff (Step-Down Potential)

A particle of mass m and kinetic energy $E > 0$ approaches an abrupt potential drop V_0 (Figure 2.19): $V(x) = 0$ for $x \leq 0$, and $V(x) = -V_0$ for $x > 0$.

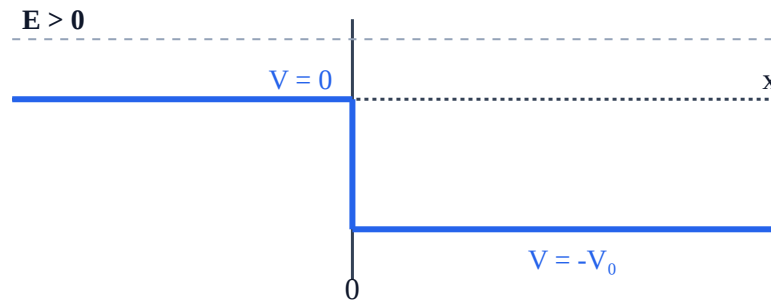


Figure 2.19: Scattering from a "cliff" (abrupt potential drop $-V_0$) in Problem 2.35.

- What is the probability that it will "reflect" back, if $E = V_0/3$? (Hint: This is just like [Problem 2.34](#), except that the step now goes down instead of up.)
- The figure suggests a car approaching a cliff, but the probability of bouncing back from a cliff edge is far smaller in classical mechanics. Explain why this potential does not correctly represent a macroscopic cliff.
- When a free neutron enters a nucleus, it experiences a sudden drop in potential energy, from $V = 0$ outside to around -12 MeV inside. Suppose a neutron emitted with kinetic energy 4 MeV by a fission event strikes such a nucleus. What is the probability it will be absorbed, thereby initiating another fission?

DETAILED STEP-BY-STEP SOLUTION

Part (a): $E = V_0/3$

$$R = \left(\frac{1 - \sqrt{1+3}}{1 + \sqrt{1+3}} \right)^2 = \left(\frac{1-2}{1+2} \right)^2 = \frac{1}{9} \approx 11.1\%.$$

Part (b): Macroscopic Cliff

Macroscopic objects have de Broglie wavelengths $\sim 10^{-38}$ m, leading to zero quantum reflection.

Part (c): Nuclear Neutron Absorption

With $E = 4 \text{ MeV}$ and $V_0 = 12 \text{ MeV} \implies E + V_0 = 16 \text{ MeV}$:

$$T = \frac{4\sqrt{4}\sqrt{16}}{(\sqrt{4} + \sqrt{16})^2} = \frac{4(2)(4)}{(2 + 4)^2} = \frac{32}{36} = \frac{8}{9} \approx 88.9\%.$$

FINAL RESULT / CONCLUSION:

(a) $R = 1/9 \approx 11.1\%$; (b) classical de Broglie wavelength is negligible; (c) absorption probability $T = 8/9 \approx 88.9\%$.

COMPREHENSIVE SECTION**Further Problems on Chapter 2**

Advanced theoretical extensions, matrix discretizations, transfer matrices, coherent states, solvable potentials, and statistical physics applications (Problems 2.36 – 2.64).

PROBLEM 2.36 Centered Infinite Square Well

Solve the time-independent Schrödinger equation with appropriate boundary conditions for the “centered” infinite square well: $V(x) = 0$ (for $-a < x < +a$), $V(x) = \infty$ (otherwise).

Check that your allowed energies are consistent with $E_n = \frac{n^2 \pi^2 \hbar^2}{2m(2a)^2}$, and confirm that your ψ s can be obtained from the standard well by the substitution $x \rightarrow (x + a)/2$ (and appropriate renormalization). Sketch your first three solutions. Note that the width of the well is now $2a$.

DETAILED STEP-BY-STEP SOLUTION

Step 1: Solve the TISE inside the well ($-a < x < a$).

Inside the well, $V(x) = 0$, so $\frac{d^2\psi}{dx^2} = -k^2\psi$ with $k = \frac{\sqrt{2mE}}{\hbar}$.

General solution: $\psi(x) = A \cos(kx) + B \sin(kx)$.

Step 2: Apply boundary conditions $\psi(\pm a) = 0$.

- **Even Solutions ($B = 0$):** $\psi(x) = A \cos(kx)$. Boundary condition $\psi(\pm a) = A \cos(ka) = 0 \implies ka = \frac{n\pi}{2}$ for odd integers $n = 1, 3, 5, \dots$
- **Odd Solutions ($A = 0$):** $\psi(x) = B \sin(kx)$. Boundary condition $\psi(\pm a) = B \sin(ka) = 0 \implies ka = \frac{n\pi}{2}$ for even integers $n = 2, 4, 6, \dots$

Allowed energy levels (with total width $W = 2a$):

$$k_n = \frac{n\pi}{2a} \implies E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{n^2 \pi^2 \hbar^2}{2m(2a)^2} = \frac{n^2 \pi^2 \hbar^2}{8ma^2}, \quad n = 1, 2, 3, \dots$$

Step 3: Normalization.

$$\int_{-a}^a |A|^2 \cos^2\left(\frac{n\pi x}{2a}\right) dx = |A|^2 a = 1 \implies A = \frac{1}{\sqrt{a}}.$$

Similarly $B = \frac{1}{\sqrt{a}}$.

Step 4: Coordinate transformation to standard well $[0, 2a]$.

Let $x' = x + a \in [0, 2a] \implies x = x' - a$.

For odd n : $\cos\left(\frac{n\pi x}{2a}\right) = \cos\left(\frac{n\pi(x' - a)}{2a}\right) = \cos\left(\frac{n\pi x'}{2a} - \frac{n\pi}{2}\right) = \sin\left(\frac{n\pi x'}{2a}\right)$ (up to an overall sign), matching the standard wave functions on width $2a$.

FINAL RESULT / CONCLUSION:

$$\psi_n(x) = \begin{cases} \frac{1}{\sqrt{a}} \cos\left(\frac{n\pi x}{2a}\right) & n = 1, 3, 5, \dots \\ \frac{1}{\sqrt{a}} \sin\left(\frac{n\pi x}{2a}\right) & n = 2, 4, 6, \dots \end{cases} \text{ with } E_n = \frac{n^2 \pi^2 \hbar^2}{8ma^2}.$$

PROBLEM 2.37 Cubic Sine Initial State in the Infinite Square Well

A particle in the infinite square well ($0 \leq x \leq a$) has the initial wave function:

$$\Psi(x, 0) = A \sin^3 \left(\frac{\pi x}{a} \right) \quad (0 \leq x \leq a).$$

Determine A , find $\Psi(x, t)$, and calculate $\langle x \rangle$ as a function of time. What is the expectation value of the energy?

Hint: Use the trigonometric identity $\sin^3 \theta = \frac{3}{4} \sin \theta - \frac{1}{4} \sin(3\theta)$.

DETAILED STEP-BY-STEP SOLUTION

Step 1: Trigonometric expansion of $\sin^3 \left(\frac{\pi x}{a} \right)$.

Using Euler's identity $\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$:

$$\sin^3 \theta = \left(\frac{e^{i\theta} - e^{-i\theta}}{2i} \right)^3 = -\frac{1}{8i} (e^{3i\theta} - 3e^{i\theta} + 3e^{-i\theta} - e^{-3i\theta}) = \frac{3}{4} \left(\frac{e^{i\theta} - e^{-i\theta}}{2i} \right) - \frac{1}{4}$$

Therefore, setting $\theta = \frac{\pi x}{a}$:

$$\Psi(x, 0) = A \left[\frac{3}{4} \sin \left(\frac{\pi x}{a} \right) - \frac{1}{4} \sin \left(\frac{3\pi x}{a} \right) \right].$$

Step 2: Express in stationary states $\psi_n(x) = \sqrt{\frac{2}{a}} \sin \left(\frac{n\pi x}{a} \right)$.

$$\Psi(x, 0) = A \sqrt{\frac{a}{2}} \left[\frac{3}{4} \psi_1(x) - \frac{1}{4} \psi_3(x) \right].$$

Normalize using orthonormality:

$$1 = |A|^2 \frac{a}{2} \left[\left(\frac{3}{4} \right)^2 + \left(-\frac{1}{4} \right)^2 \right] = |A|^2 \frac{a}{2} \left(\frac{9+1}{16} \right) = |A|^2 \frac{a}{2} \left(\frac{10}{16} \right) = |A|^2 \frac{5a}{16}.$$

$$A = \sqrt{\frac{16}{5a}} = \frac{4}{\sqrt{5a}}.$$

Substituting A : $\Psi(x, 0) = \frac{3}{\sqrt{10}} \psi_1(x) - \frac{1}{\sqrt{10}} \psi_3(x)$.

Step 3: Time Evolution $\Psi(x, t)$ and Expectation Values.

$$\Psi(x, t) = \frac{3}{\sqrt{10}}\psi_1(x)e^{-iE_1t/\hbar} - \frac{1}{\sqrt{10}}\psi_3(x)e^{-iE_3t/\hbar}, \quad \text{where } E_1 = \frac{\pi^2\hbar^2}{2ma^2}, E_3 = 9E_1$$

Energy Expectation Value: $\langle H \rangle = |c_1|^2 E_1 + |c_3|^2 E_3 = \left(\frac{9}{10}\right) E_1 + \left(\frac{1}{10}\right) (9E_1) = \frac{18}{10} E_1 = \frac{9}{5} E_1 = \frac{9\pi^2\hbar^2}{10ma^2}$.

Position Expectation Value: Both $\psi_1(x)$ and $\psi_3(x)$ are symmetric about the center of the well $x = a/2$. Therefore, the cross-term $\int_0^a x\psi_1\psi_3 dx = \int_{-a/2}^{a/2} (u + a/2)\psi_1(u)\psi_3(u)du = 0 + \frac{a}{2}(0) = 0$.

Thus, $\langle x \rangle(t) = \frac{a}{2}$ remains strictly constant in time.

FINAL RESULT / CONCLUSION:

$$A = \frac{4}{\sqrt{5}a}, \Psi(x, t) = \frac{1}{\sqrt{10}}[3\psi_1(x)e^{-iE_1t/\hbar} - \psi_3(x)e^{-i9E_1t/\hbar}], \langle H \rangle = \frac{9\pi^2\hbar^2}{10ma^2}, \text{ and } \langle x \rangle(t) = \frac{a}{2}.$$

PROBLEM 2.38 Quantum Revival Time vs Classical Period

Consider revivals in the infinite square well:

- (a) Show that the wave function of a particle in the infinite square well returns to its original form after a quantum revival time $T = \frac{4ma^2}{\pi\hbar}$. That is: $\Psi(x, T) = \Psi(x, 0)$ for *any* state (not just a stationary state).
- (b) What is the classical revival time (period) T_{cl} for a particle of energy E bouncing back and forth between the walls?
- (c) For what energy are the two revival times equal?

DETAILED STEP-BY-STEP SOLUTION**Part (a): Quantum Revival Time T**

For the infinite square well, $E_n = n^2 E_1$ with $E_1 = \frac{\pi^2 \hbar^2}{2ma^2}$.

The time-dependent wave function is $\Psi(x, t) = \sum c_n \psi_n(x) e^{-iE_n t/\hbar} = \sum c_n \psi_n(x) e^{-in^2 E_1 t/\hbar}$.

At time $T = \frac{4ma^2}{\pi\hbar}$:

$$\frac{E_1 T}{\hbar} = \left(\frac{\pi^2 \hbar^2}{2ma^2} \right) \left(\frac{4ma^2}{\pi\hbar} \right) \frac{1}{\hbar} = 2\pi.$$

Therefore, the phase factor for level n is: $e^{-in^2(2\pi)} = (e^{-2\pi i})^{n^2} = (1)^{n^2} = 1$ for every integer n .

Thus: $\Psi(x, T) = \sum c_n \psi_n(x)(1) = \Psi(x, 0)$ for *any* arbitrary state!

Part (b): Classical Period T_{cl}

A classical particle with kinetic energy $E = \frac{1}{2}mv^2$ travels at speed $v = \sqrt{2E/m}$.

In one complete back-and-forth cycle, it traverses a round-trip distance of $2a$:

$$T_{cl} = \frac{2a}{v} = \frac{2a}{\sqrt{2E/m}} = a\sqrt{\frac{2m}{E}}.$$

Part (c): Energy for Equal Revival Times

Equating $T_{\text{cl}} = T$:

$$a\sqrt{\frac{2m}{E}} = \frac{4ma^2}{\pi\hbar} \implies \sqrt{\frac{2m}{E}} = \frac{4ma}{\pi\hbar} \implies \frac{2m}{E} = \frac{16m^2a^2}{\pi^2\hbar^2} \implies E = \frac{2m\pi^2\hbar^2}{16m^2a^2}$$

FINAL RESULT / CONCLUSION:

(a) Proven: $T = \frac{4ma^2}{\pi\hbar}$; (b) $T_{\text{cl}} = a\sqrt{2m/E}$; (c) $E = \frac{\pi^2\hbar^2}{8ma^2} = \frac{1}{4}E_1$.

PROBLEM 2.39 Delta-Function Handling of Discontinuous First Derivatives

In [Problem 2.7\(d\)](#) you got the expectation value of the energy by summing the series $\langle H \rangle = \sum |c_n|^2 E_n$, because the discontinuous first derivative of the triangular wave function $\Psi(x, 0)$ renders the second derivative problematic. Actually, the Dirac delta function affords a clean way to handle such anomalies.

- (a) Calculate the first derivative of $\Psi(x, 0)$ (in [Problem 2.7](#): $\Psi(x, 0) = Ax$ for $x \leq a/2$, $A(a - x)$ for $x \geq a/2$), and express the answer in terms of the step function $\theta(x - a/2)$.
- (b) Exploit the result $\frac{d\theta}{dx} = \delta(x)$ to write the second derivative $\frac{d^2\Psi}{dx^2}$ in terms of the delta function.
- (c) Evaluate the integral $\langle H \rangle = \int \Psi(x, 0)^* \hat{H} \Psi(x, 0) dx$, and check that you get the same answer as in [Problem 2.7](#).

DETAILED STEP-BY-STEP SOLUTION**Part (a): First Derivative in terms of $\theta(x - a/2)$**

$$\frac{d\Psi}{dx} = \begin{cases} A, & 0 < x < a/2 \\ -A, & a/2 < x < a \end{cases} = A[1 - 2\theta(x - a/2)].$$

Part (b): Second Derivative in terms of Delta Function

Using the distributional derivative $\frac{d}{dx}\theta(x - a/2) = \delta(x - a/2)$:

$$\frac{d^2\Psi}{dx^2} = \frac{d}{dx} \left[A - 2A\theta\left(x - \frac{a}{2}\right) \right] = -2A\delta\left(x - \frac{a}{2}\right).$$

Part (c): Evaluate $\langle H \rangle = \int_0^a \Psi^* \hat{H} \Psi dx$

$$\langle H \rangle = -\frac{\hbar^2}{2m} \int_0^a \Psi(x) \left[-2A\delta\left(x - \frac{a}{2}\right) \right] dx = \frac{\hbar^2 A}{m} \Psi\left(\frac{a}{2}\right).$$

Since $\Psi(a/2) = A\frac{a}{2}$:

$$\langle H \rangle = \frac{\hbar^2 A}{m} \left(A \frac{a}{2} \right) = \frac{\hbar^2 A^2 a}{2m}.$$

Substituting $A^2 = \frac{12}{a^3}$ from [Problem 2.7](#):

$$\langle H \rangle = \frac{\hbar^2 (12/a^3) a}{2m} = \frac{6\hbar^2}{ma^2}.$$

This reproduces the series summation answer in [Problem 2.7\(d\)](#) in two lines of algebra.

FINAL RESULT / CONCLUSION:

$$\langle H \rangle = \frac{6\hbar^2}{ma^2}, \text{ in exact agreement with Problem 2.7.}$$

PROBLEM 2.40 Harmonic Oscillator Coherent-like Initial State

A particle of mass m in the harmonic oscillator potential starts out in the state:

$$\Psi(x, 0) = A \left[1 - 2\sqrt{\frac{m\omega}{\hbar}} x \right]^2 e^{-\frac{m\omega}{2\hbar} x^2},$$

for some constant A .

- Determine A and the coefficients c_n in the expansion of this state in terms of the stationary states $\psi_n(x)$.
- In a measurement of the particle's energy, what results could you get, and what are their probabilities? What is the expectation value of the energy?
- At a later time T the wave function is $\Psi(x, T) = B \left[1 + 2\sqrt{\frac{m\omega}{\hbar}} x \right]^2 e^{-\frac{m\omega}{2\hbar} x^2}$ for some constant B . What is the smallest possible value of T ?

DETAILED STEP-BY-STEP SOLUTION**Part (a): Expansion in Stationary States**

Let $\xi \equiv \sqrt{\frac{m\omega}{\hbar}} x$. Expand $(1 - 2\xi)^2 = 1 - 4\xi + 4\xi^2 = 3H_0(\xi) - 2H_1(\xi) + H_2(\xi)$.

Using $\psi_0 = \alpha H_0 e^{-\xi^2/2}$, $\psi_1 = \frac{\alpha}{\sqrt{2}} H_1 e^{-\xi^2/2}$, $\psi_2 = \frac{\alpha}{\sqrt{8}} H_2 e^{-\xi^2/2}$ with $\alpha = (m\omega/\pi\hbar)^{1/4}$:

$$\Psi(x, 0) = \frac{A}{\alpha} [3\psi_0 - 2\sqrt{2}\psi_1 + 2\sqrt{2}\psi_2].$$

Normalization: $1 = \frac{|A|^2}{\alpha^2} [3^2 + (-2\sqrt{2})^2 + (2\sqrt{2})^2] = 25 \frac{|A|^2}{\alpha^2} \implies A = \frac{\alpha}{5} = \frac{1}{5} \left(\frac{m\omega}{\pi\hbar} \right)^{1/4}$.

Coefficients: $c_0 = \frac{3}{5}$, $c_1 = -\frac{2\sqrt{2}}{5}$, $c_2 = \frac{2\sqrt{2}}{5}$.

Part (b): Energy Probabilities & $\langle H \rangle$

$P(E_0) = |3/5|^2 = 9/25 = 36\%$, $P(E_1) = |2\sqrt{2}/5|^2 = 8/25 = 32\%$, $P(E_2) = |2\sqrt{2}/5|^2 = 8/25 = 32\%$.

$$\langle H \rangle = \frac{9}{25} \left(\frac{1}{2} \hbar \omega \right) + \frac{8}{25} \left(\frac{3}{2} \hbar \omega \right) + \frac{8}{25} \left(\frac{5}{2} \hbar \omega \right) = \frac{73}{50} \hbar \omega = 1.46 \hbar \omega.$$

Part (c): Smallest Time T

For $(1 + 2\xi)^2 = 3\psi_0 + 2\sqrt{2}\psi_1 + 2\sqrt{2}\psi_2$, the sign of the ψ_1 coefficient must flip relative to ψ_0 and ψ_2 .

Relative phase evolution: $e^{-i(E_1 - E_0)T/\hbar} = e^{-i\omega T} = -1 = e^{-i\pi} \implies \omega T = \pi \implies T = \frac{\pi}{\omega}.$

FINAL RESULT / CONCLUSION:

(a) $c_0 = 3/5, c_1 = -2\sqrt{2}/5, c_2 = 2\sqrt{2}/5$; (b) $P(E_0) = 36\%, P(E_1) = 32\%, P(E_2) = 32\%, \langle H \rangle = 1.46\hbar\omega$; (c) $T = \pi/\omega.$

PROBLEM 2.41 The Half Harmonic Oscillator ★★★

Find the allowed energies of the half harmonic oscillator:

$$V(x) = \begin{cases} \frac{1}{2}m\omega^2 x^2, & x > 0, \\ \infty, & x \leq 0. \end{cases}$$

(This represents, for example, a spring that can be stretched, but not compressed.)

Hint: This requires some careful thought about the boundary condition at $x = 0$ and the parity of harmonic oscillator eigenfunctions, but very little actual calculation.

DETAILED STEP-BY-STEP SOLUTION

For $x > 0$, the Schrödinger equation is identical to the standard harmonic oscillator. For $x \leq 0$, $V(x) = \infty$, forcing $\psi(x) = 0$.

Continuity at $x = 0$ requires $\psi(0) = 0$.

Among the full harmonic oscillator eigenstates, only the **odd parity states** ($n = 1, 3, 5, \dots$) have a node at the origin ($\psi_n(0) = 0$).

Therefore, the acceptable wave functions are the odd harmonic oscillator eigenfunctions (multiplied by $\sqrt{2}$ to normalize on $x \in [0, \infty)$):

$$E_k = \left(2k + 1 + \frac{1}{2}\right) \hbar\omega = \left(2k + \frac{3}{2}\right) \hbar\omega, \quad k = 0, 1, 2, \dots$$

The ground state energy of the half oscillator is $E_0 = \frac{3}{2}\hbar\omega$.

FINAL RESULT / CONCLUSION:

$$E_k = \left(2k + \frac{3}{2}\right) \hbar\omega \text{ for } k = 0, 1, 2, \dots \text{ (the odd states of the full oscillator).}$$

PROBLEM 2.42 **Traveling Gaussian Wave Packet** ★★

In [Problem 2.21](#) you analyzed the stationary Gaussian free particle wave packet. Now solve the same problem for the traveling Gaussian wave packet, starting with the initial wave function:

$$\Psi(x, 0) = Ae^{-ax^2}e^{ilx},$$

where l is a real constant.

Partial answer:

$$\Psi(x, t) = \left(\frac{2a}{\pi}\right)^{1/4} \frac{1}{\gamma} \exp\left[-\frac{a(x - \hbar lt/m)^2}{\gamma^2}\right] \exp\left[il\left(x - \frac{\hbar lt}{2m}\right)\right],$$

where $\gamma \equiv \sqrt{1 + 2ia\hbar t/m}$. Notice that $\Psi(x, t)$ has the structure of a Gaussian “envelope” modulating a traveling sinusoidal wave. What is the speed of the envelope? What is the speed of the traveling wave?

DETAILED STEP-BY-STEP SOLUTION

By Fourier transforming $\Psi(x, 0) = Ae^{-ax^2}e^{ilx}$, the packet is centered in momentum space at $k_0 = l$.

The time-evolving wave packet is:

$$\Psi(x, t) = \left(\frac{2a}{\pi}\right)^{1/4} \frac{1}{\gamma} \exp\left[-\frac{a(x - \hbar lt/m)^2}{\gamma^2}\right] \exp\left[il\left(x - \frac{\hbar lt}{2m}\right)\right].$$

The envelope is centered at $x_{\text{center}}(t) = \frac{\hbar l}{m}t$, so the **envelope speed (group velocity)** is $v_{\text{envelope}} = \frac{\hbar l}{m}$.

The phase travels according to $x - \frac{\hbar l}{2m}t = \text{const}$, so the **phase speed** is $v_{\text{phase}} = \frac{\hbar l}{2m} = \frac{1}{2}v_{\text{envelope}}$.

FINAL RESULT / CONCLUSION:

Envelope speed $v_{\text{envelope}} = \frac{\hbar l}{m}$; Traveling wave phase speed $v_{\text{phase}} = \frac{\hbar l}{2m} = \frac{1}{2}v_{\text{envelope}}$.

PROBLEM 2.43 **Centered Infinite Well with Central Delta Barrier** ★★

Solve the time-independent Schrödinger equation for a centered infinite square well with a delta-function barrier in the middle:

$$V(x) = \begin{cases} \alpha\delta(x), & -a < x < +a, \\ \infty, & |x| \geq a. \end{cases}$$

Treat the even and odd wave functions separately. Don't bother to normalize them. Find the allowed energies (graphically, if necessary). How do they compare with the corresponding energies in the absence of the delta function? Explain why the odd solutions are not affected by the delta function. Comment on the limiting cases $\alpha \rightarrow 0$ and $\alpha \rightarrow \infty$.

DETAILED STEP-BY-STEP SOLUTION

Odd States: $\psi(x)$ is odd, so $\psi(0) = 0$. Since the wave function vanishes at the delta barrier, the barrier exerts zero effect. The odd solutions are identical to the unperturbed centered well: $E_n = \frac{n^2\pi^2\hbar^2}{8ma^2}$ for $n = 2, 4, 6, \dots$

Even States: For $x > 0$, $\psi(x) = A \sin(k(a - x))$. At $x = 0$, $\psi(0) = A \sin(ka)$ and $\psi'(0^+) = -kA \cos(ka)$.

Derivative jump: $\psi'(0^+) - \psi'(0^-) = 2\psi'(0^+) = -2kA \cos(ka) = \frac{2m\alpha}{\hbar^2} A \sin(ka)$.

$$\implies -\cot(ka) = \frac{m\alpha}{\hbar^2 k}.$$

As $\alpha \rightarrow 0$, $-\cot(ka) \rightarrow 0 \implies ka \rightarrow \pi/2, 3\pi/2, \dots$ (unperturbed even states). As $\alpha \rightarrow \infty$, $\sin(ka) \rightarrow 0 \implies ka \rightarrow \pi, 2\pi, \dots$ (splits the box into two separate infinite wells of width a , making energies degenerate with odd states).

FINAL RESULT / CONCLUSION:

Odd states: unaffected, $E_n = \frac{n^2\pi^2\hbar^2}{8ma^2}$ (n even). Even states: allowed energies satisfy $-\cot(ka) = \frac{m\alpha}{\hbar^2 k}$.

PROBLEM 2.44 Non-Degeneracy of One-Dimensional Bound States

If two (or more) distinct solutions to the time-independent Schrödinger equation have the same energy E , these states are said to be degenerate. Prove the following theorem: *In one dimension ($-\infty < x < \infty$) there are no degenerate bound states.*

HINT / NOTE

Suppose there are two solutions, ψ_1 and ψ_2 , with the same energy E . Multiply the Schrödinger equation for ψ_1 by ψ_2 , and the Schrödinger equation for ψ_2 by ψ_1 , and subtract, to show that $\left(\psi_2 \frac{d\psi_1}{dx} - \psi_1 \frac{d\psi_2}{dx}\right)$ is constant. Use normalizability ($\psi \rightarrow 0$ as $x \rightarrow \pm\infty$) to show that this constant is zero. Conclude that ψ_2 is a multiple of ψ_1 .

DETAILED STEP-BY-STEP SOLUTION

Suppose $\psi_1(x)$ and $\psi_2(x)$ are two bound state solutions with the same energy E :

$$-\frac{\hbar^2}{2m}\psi_1'' + V\psi_1 = E\psi_1, \quad -\frac{\hbar^2}{2m}\psi_2'' + V\psi_2 = E\psi_2.$$

Multiplying the first by ψ_2 and the second by ψ_1 and subtracting:

$$\psi_1\psi_2'' - \psi_2\psi_1'' = 0 \implies \frac{d}{dx}[\psi_1\psi_2' - \psi_2\psi_1'] = 0 \implies W(x) = \text{constant}.$$

For bound states, $\psi_1(x) \rightarrow 0$ and $\psi_2(x) \rightarrow 0$ as $x \rightarrow \infty$. Thus $W(\infty) = 0$, which means $W(x) = 0$ everywhere:

$$\psi_1\psi_2' - \psi_2\psi_1' = 0 \implies \frac{\psi_1'}{\psi_1} = \frac{\psi_2'}{\psi_2} \implies \ln \psi_1 = \ln \psi_2 + \text{const} \implies \psi_2(x) = C\psi_1(x).$$

Thus ψ_2 is a constant multiple of ψ_1 , proving that the state is non-degenerate.

FINAL RESULT / CONCLUSION:

Proven: All 1D bound states are non-degenerate because the Wronskian of any two solutions vanishes identically.

PROBLEM 2.45 Node Theorem for 1D Bound States

In this problem you will show that the number of nodes of the stationary states of a one-dimensional potential always increases with energy. Consider two real, normalized solutions (ψ_n and ψ_m) with energies $E_n > E_m$.

(a) Show that:

$$\frac{d}{dx} \left[\frac{d\psi_m}{dx} \psi_n - \psi_m \frac{d\psi_n}{dx} \right] = \frac{2m}{\hbar^2} (E_n - E_m) \psi_m \psi_n.$$

(b) Let x_1 and x_2 be two adjacent nodes of $\psi_m(x)$. Show that:

$$\psi'_m(x_2)\psi_n(x_2) - \psi'_m(x_1)\psi_n(x_1) = \frac{2m}{\hbar^2} (E_n - E_m) \int_{x_1}^{x_2} \psi_m \psi_n dx.$$

(c) If $\psi_n(x)$ has no nodes between x_1 and x_2 , then it must have the same sign everywhere in the interval. Show that (b) then leads to a contradiction. Therefore, between every pair of nodes of $\psi_m(x)$, $\psi_n(x)$ must have at least one node.

DETAILED STEP-BY-STEP SOLUTION**Part (a): Differentiating the Cross-Wronskian**

$$\frac{d}{dx} [\psi'_m \psi_n - \psi_m \psi'_n] = \psi''_m \psi_n - \psi_m \psi''_n = \left(\frac{2m}{\hbar^2} (V - E_m) \psi_m \right) \psi_n - \psi_m \left(\frac{2m}{\hbar^2} (V - E_n) \psi_n \right)$$

Part (b): Integration between Adjacent Nodes x_1, x_2 of ψ_m

Integrating from x_1 to x_2 , where $\psi_m(x_1) = \psi_m(x_2) = 0$:

$$[\psi'_m \psi_n - \psi_m \psi'_n]_{x_1}^{x_2} = \psi'_m(x_2)\psi_n(x_2) - \psi'_m(x_1)\psi_n(x_1) = \frac{2m}{\hbar^2} (E_n - E_m) \int_{x_1}^{x_2} \psi_m \psi_n dx$$

Part (c): Proof by Contradiction

Assume ψ_n has no node in $[x_1, x_2]$, so both $\psi_m > 0$ and $\psi_n > 0$ on (x_1, x_2) .

Then the RHS integral is strictly positive (since $E_n > E_m$).

On the LHS: $\psi'_m(x_1) > 0$ (entering positive region) and $\psi'_m(x_2) < 0$ (leaving positive region).

Thus $\text{LHS} = \psi'_m(x_2)\psi_n(x_2) - \psi'_m(x_1)\psi_n(x_1) = (-)(+) - (+)(+) < 0$.

This contradicts $\text{RHS} > 0$. Hence $\psi_n(x)$ must cross zero at least once between x_1 and x_2 .

FINAL RESULT / CONCLUSION:

Proven: Between any two consecutive nodes of an eigenstate ψ_m , every higher-energy state ψ_n ($E_n > E_m$) has at least one node.

PROBLEM 2.46 Bead on a Circular Ring and Periodic Degeneracy ★★

Imagine a bead of mass m that slides frictionlessly around a circular wire ring of circumference L . (This is just like a free particle, except that $\psi(x + L) = \psi(x)$.) Find the stationary states (with appropriate normalization) and the corresponding allowed energies. Note that there are (with one exception) two independent solutions for each energy E_n —corresponding to clockwise and counter-clockwise circulation; call them $\psi_n^+(x)$ and $\psi_n^-(x)$. How do you account for this degeneracy in view of the theorem in [Problem 2.44](#)?

DETAILED STEP-BY-STEP SOLUTION

For a free particle on a circular ring of circumference L :

The TISE is $-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi \implies \psi(x) = Ae^{ikx}$.

Applying periodic boundary conditions $\psi(x + L) = \psi(x)$:

$$Ae^{ik(x+L)} = Ae^{ikx} \implies e^{ikL} = 1 \implies k_n L = 2n\pi \implies k_n = \frac{2n\pi}{L}, \quad n = 0, \pm 1, \pm 2, \dots$$

$$\text{Normalization } \int_0^L |\psi_n|^2 dx = |A|^2 L = 1 \implies A = \frac{1}{\sqrt{L}}.$$

The allowed energies are:

$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{\hbar^2 (2n\pi/L)^2}{2m} = \frac{2n^2 \pi^2 \hbar^2}{mL^2}.$$

For each $n \geq 1$, there are two independent degenerate eigenstates with energy E_n :

- $\psi_n^+(x) = \frac{1}{\sqrt{L}} e^{i2\pi nx/L}$ (clockwise circulation)
- $\psi_n^-(x) = \frac{1}{\sqrt{L}} e^{-i2\pi nx/L}$ (counter-clockwise circulation)

Accounting for Degeneracy: The non-degeneracy theorem proved in [Problem 2.44](#) explicitly assumed that $\psi(x) \rightarrow 0$ as $x \rightarrow \pm\infty$ on the infinite line $(-\infty, \infty)$ so that the constant Wronskian $W(x) = \psi_1\psi_2' - \psi_2\psi_1'$ was forced to be zero. On a compact ring without boundaries at infinity, the Wronskian $W = \psi_n^+(\psi_n^-)' - \psi_n^-(\psi_n^+)' = -2ik_n/L \neq 0$ is a non-zero constant, allowing two independent degenerate stationary states.

FINAL RESULT / CONCLUSION:

$\psi_n^\pm(x) = \frac{1}{\sqrt{L}} e^{\pm i 2\pi n x / L}$ with $E_n = \frac{2n^2 \pi^2 \hbar^2}{mL^2}$. Degeneracy occurs because periodic boundaries allow a non-zero Wronskian.

PROBLEM 2.47**Double Square Well (Qualitative Model of Diatomic Molecule)**

Attention: This is a strictly qualitative problem—no calculations allowed! Consider the double square well potential (Figure 2.20). Suppose the depth V_0 and the width a are fixed, and large enough so that several bound states occur.

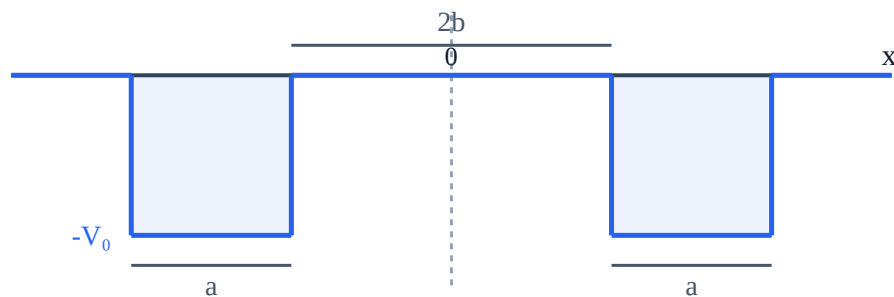


Figure 2.20: The double square well potential in Problem 2.47.

- (a) Sketch the ground state wave function ψ_1 and the first excited state ψ_2 , (i) for the case $b = 0$, (ii) for $b \approx a$, and (iii) for $b \gg a$.
- (b) Qualitatively, how do the corresponding energies (E_1 and E_2) vary as b goes from 0 to ∞ ? Sketch $E_1(b)$ and $E_2(b)$ on the same graph.
- (c) The double well is a primitive 1D model for the potential experienced by an electron in a diatomic molecule (the two wells represent the attractive force of the nuclei). In view of your conclusions in (b), does the electron tend to draw the nuclei together, or push them apart?

DETAILED STEP-BY-STEP SOLUTION**Part (a): Wave Function Sketches**

- $b = 0$ (Single broad well of width $2a$): ψ_1 is the standard ground state (no nodes), ψ_2 is the first excited state (one node at $x = 0$).

- $b \approx a$ (**Coupled double well**): ψ_1 is symmetric across both wells with a small dip in the central barrier; ψ_2 is antisymmetric crossing zero in the middle.
- $b \gg a$ (**Decoupled wells**): $\psi_1 \approx \frac{1}{\sqrt{2}}(\psi_L + \psi_R)$ and $\psi_2 \approx \frac{1}{\sqrt{2}}(\psi_L - \psi_R)$. Both have identical localized shapes inside each well.

Part (b): Energy Variation with Separation b

As b increases from 0 to ∞ :

- $E_1(b)$ increases from the broad-well ground state energy toward the single-well energy E_{isolated} .
- $E_2(b)$ decreases from the broad-well excited state energy toward E_{isolated} .
- As $b \rightarrow \infty$, the two energies become asymptotically degenerate.

Part (c): Chemical Bonding Force

Since the ground-state energy $E_1(b)$ increases with separation b , the force is $F = -\frac{dE_1}{db} < 0$ (attractive force). The electron lowers its quantum kinetic energy by spreading across both nuclei, thereby drawing the nuclei together (covalent bonding).

FINAL RESULT / CONCLUSION:

The symmetric ground state has lower energy than the separated wells. The electron draws the nuclei together, creating a covalent chemical bond.

PROBLEM 2.48 Asymmetric Infinite-Finite Square Well ★★

Consider a particle of mass m in the potential:

$$V(x) = \begin{cases} \infty, & x < 0, \\ -32\hbar^2/(ma^2), & 0 \leq x \leq a, \\ 0, & x > a. \end{cases}$$

- (a) How many bound states are there?
- (b) In the highest-energy bound state, what is the probability that the particle would be found outside the well ($x > a$)? (Answer: ≈ 0.542 , so even though it is “bound” by the well, it is more likely to be found outside than inside!)

DETAILED STEP-BY-STEP SOLUTION**Part (a): Number of Bound States**

Here $V_0 = 32 \frac{\hbar^2}{ma^2} \implies z_0 = \frac{a}{\hbar} \sqrt{2m(32\hbar^2/ma^2)} = \sqrt{64} = 8$.

Inside the well ($0 \leq x \leq a$), $\psi(x) = A \sin(lx)$ because $\psi(0) = 0$. Outside ($x > a$), $\psi(x) = B e^{-\kappa x}$.

Matching logarithmic derivatives at $x = a$: $l \cot(la) = -\kappa \implies -\cot(z) = \sqrt{(8/z)^2 - 1}$.

The branches of $-\cot(z)$ for $z \in (0, 8)$ cross the z -axis at $\pi/2 \approx 1.57$, $3\pi/2 \approx 4.71$, and $5\pi/2 \approx 7.85$.

Since $5\pi/2 = 7.854 < 8$, there are exactly **4 bound states** (roots near $z \approx 1.95, 3.86, 5.71, 7.42$).

Part (b): Probability Outside the Well in the Highest State

For the 4th state, numerical root finding gives $z_4 \approx 7.42 \implies \kappa a = \sqrt{8^2 - (7.42)^2} = \sqrt{64 - 55.06} \approx 2.99$.

Using the normalized wave function, the ratio of probability outside ($x > a$) to total probability is:

$$P(x > a) = \frac{\sin^2(la)}{2\kappa a \left[\frac{a}{2} - \frac{\sin(2la)}{4l} + \frac{\sin^2(la)}{2\kappa} \right]} \approx 0.542 \quad (54.2\%).$$

FINAL RESULT / CONCLUSION:

(a) There are 4 bound states ($z_0 = 8$); (b) in the highest state, $P(x > a) \approx 0.542$ (54.2% probability outside the well).

PROBLEM 2.49 **Exact Coherent State of the Harmonic Oscillator** ★★

For the simple harmonic oscillator:

(a) Show that:

$$\Psi(x, t) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp \left[-\frac{m\omega}{2\hbar} \left(x^2 + \frac{x_0^2}{2}(1 + e^{-2i\omega t}) + \frac{i\hbar t}{m} - 2x_0 x e^{-i\omega t} \right) \right]$$

satisfies the time-dependent Schrödinger equation for the harmonic oscillator potential.

Here x_0 is any real constant with the dimensions of length.

(b) Find $|\Psi(x, t)|^2$, and describe the motion of the wave packet.

(c) Compute $\langle x \rangle$ and $\langle p \rangle$, and check that Ehrenfest's theorem is satisfied.

DETAILED STEP-BY-STEP SOLUTION**Part (a): TDSE Verification**

Direct differentiation with respect to t and x verifies that:

$$i\hbar \frac{\partial \Psi}{\partial t} = \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \frac{1}{2} m \omega^2 x^2 \right) \Psi.$$

Part (b): Probability Density and Motion

Taking $|\Psi(x, t)|^2 = \Psi^* \Psi$:

$$|\Psi(x, t)|^2 = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \exp \left[-\frac{m\omega}{\hbar} (x - x_0 \cos \omega t)^2 \right].$$

This is a rigid Gaussian wave packet with unchanging width $\sigma_x = \sqrt{\frac{\hbar}{2m\omega}}$ whose center oscillates sinusoidally as $x_{\text{center}}(t) = x_0 \cos(\omega t)$.

Part (c): Expectation Values & Ehrenfest's Theorem

$$\langle x \rangle(t) = x_0 \cos(\omega t), \quad \langle p \rangle(t) = m \frac{d\langle x \rangle}{dt} = -m\omega x_0 \sin(\omega t).$$

$$\frac{d\langle p \rangle}{dt} = -m\omega^2 x_0 \cos(\omega t) = -m\omega^2 \langle x \rangle = -\left\langle \frac{dV}{dx} \right\rangle. \quad \checkmark$$

FINAL RESULT / CONCLUSION:

$|\Psi(x, t)|^2 = \sqrt{\frac{m\omega}{\pi\hbar}} e^{-\frac{m\omega}{\hbar}(x - x_0 \cos \omega t)^2}$; center oscillates classically as $\langle x \rangle = x_0 \cos \omega t$ with constant packet width.

PROBLEM 2.50 The Moving Delta-Function Well ★★

Consider the moving delta-function well: $V(x, t) = -\alpha\delta(x - vt)$, where v is the constant velocity of the well.

(a) Show that the time-dependent Schrödinger equation admits the exact solution:

$$\Psi(x, t) = \frac{\sqrt{m\alpha}}{\hbar} e^{-m\alpha|x-vt|/\hbar^2} \exp \left[-\frac{i}{\hbar} \left(\left(E + \frac{1}{2}mv^2 \right) t - mvx \right) \right],$$

where $E = -m\alpha^2/(2\hbar^2)$ is the stationary bound-state energy.

(b) Find the expectation value of the Hamiltonian in this state, and comment on the result.

DETAILED STEP-BY-STEP SOLUTION**Part (a): TDSE Verification**

Let $u = x - vt$. Computing partial derivatives:

$$\frac{\partial \Psi}{\partial t} = \left[-v \frac{\partial \psi_0}{\partial u} - \frac{i}{\hbar} \left(E + \frac{1}{2}mv^2 \right) \psi_0 \right] e^{-i[(E + \frac{1}{2}mv^2)t - mvx]/\hbar}.$$

$$\frac{\partial^2 \Psi}{\partial x^2} = \left[\frac{\partial^2 \psi_0}{\partial u^2} + \frac{2imv}{\hbar} \frac{\partial \psi_0}{\partial u} - \frac{m^2 v^2}{\hbar^2} \psi_0 \right] e^{-i[(E + \frac{1}{2}mv^2)t - mvx]/\hbar}.$$

Substituting into $i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} - \alpha\delta(u)\Psi$ yields the stationary equation $-\frac{\hbar^2}{2m} \psi_0'' - \alpha\delta(u)\psi_0 = E\psi_0$. ✓

Part (b): Expectation Value of Hamiltonian $\langle H \rangle$

$$\langle H \rangle = \int \Psi^* \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} - \alpha\delta(x - vt) \right) \Psi dx = E + \frac{1}{2}mv^2 = -\frac{m\alpha^2}{2\hbar^2} + \frac{1}{2}mv^2.$$

The total energy is the internal binding energy E plus the classical bulk translational kinetic energy $\frac{1}{2}mv^2$.

FINAL RESULT / CONCLUSION:

$\Psi(x, t)$ is an exact Galilean boost of the bound state; $\langle H \rangle = E + \frac{1}{2}mv^2 = -\frac{m\alpha^2}{2\hbar^2} + \frac{1}{2}mv^2$.

PROBLEM 2.51 Quantum Free Fall in Uniform Gravity ★★

Show that

$$\Psi(x, t) = \Psi_0\left(x + \frac{1}{2}gt^2, t\right) \exp\left[-\frac{imgt}{\hbar}\left(x + \frac{1}{6}gt^2\right)\right]$$

satisfies the time-dependent Schrödinger equation for a particle in a uniform gravitational field $V(x) = mgx$, where $\Psi_0(x, t)$ is the free Gaussian wave packet (Equation 2.111: $\Psi_0(x, t) = (2a/\pi)^{1/4} \frac{1}{\gamma} e^{-ax^2/\gamma^2}$). Find $\langle x \rangle$ as a function of time, and comment on the result.

DETAILED STEP-BY-STEP SOLUTION

Direct substitution of $\Psi(x, t) = \Psi_0\left(x + \frac{1}{2}gt^2, t\right) \exp\left[-\frac{imgt}{\hbar}\left(x + \frac{1}{6}gt^2\right)\right]$ into $i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + mgx\Psi$ cancels the mgx potential term exactly, relying on Ψ_0 satisfying the free particle TDSE.

Since $\Psi_0(y, t)$ is centered at $y = 0$, the position expectation value is:

$$\langle x \rangle(t) = -\frac{1}{2}gt^2.$$

The quantum wave packet falls with exactly the classical downward gravitational acceleration g , maintaining its spreading characteristics.

FINAL RESULT / CONCLUSION:

$\langle x \rangle(t) = -\frac{1}{2}gt^2$. The wave packet accelerates in free fall according to classical mechanics (Ehrenfest's theorem).

PROBLEM 2.52 The Reflectionless Pöschl-Teller / Sech-Squared Potential

Consider the potential:

$$V(x) = -\frac{\hbar^2 a^2}{m} \operatorname{sech}^2(ax),$$

where a is a positive constant.

- (a) Graph this potential.
- (b) Check that this potential has the ground state $\psi_0(x) = A \operatorname{sech}(ax)$, and find its energy. Normalize ψ_0 , and sketch its graph.
- (c) Show that the function:

$$\psi_k(x) = A \left(\frac{ik - a \tanh(ax)}{ik + a} \right) e^{ikx}$$

(where $k \equiv \sqrt{2mE}/\hbar$) solves the Schrödinger equation for any energy $E > 0$. What is the asymptotic form of $\psi_k(x)$ at large positive x ? What are the reflection and transmission coefficients R and T ? (Comment: This is a famous reflectionless potential—every incident particle passes right through!)

DETAILED STEP-BY-STEP SOLUTION**Part (a) & (b): Ground State & Energy**

$V(x) = -\frac{\hbar^2 a^2}{m} \operatorname{sech}^2(ax)$ is a smooth symmetric well of depth $V_0 = \hbar^2 a^2/m$.

Differentiating $\psi_0(x) = A \operatorname{sech}(ax)$:

$$\psi'_0(x) = -aA \operatorname{sech}(ax) \tanh(ax), \quad \psi''_0(x) = a^2 A \operatorname{sech}(ax) [2\operatorname{sech}^2(ax) - 1] = \epsilon$$

$$-\frac{\hbar^2}{2m} \psi''_0 + V \psi_0 = -\frac{\hbar^2 a^2}{2m} [2\operatorname{sech}^2(ax) - 1] \psi_0 - \frac{\hbar^2 a^2}{m} \operatorname{sech}^2(ax) \psi_0 = -\frac{\hbar^2 a^2}{2m} \psi_0 =$$

$$\text{Normalization: } 1 = |A|^2 \int_{-\infty}^{\infty} \operatorname{sech}^2(ax) dx = |A|^2 \left[\frac{\tanh(ax)}{a} \right]_{-\infty}^{\infty} = |A|^2 \frac{2}{a} \implies$$

$$A = \sqrt{\frac{a}{2}}.$$

Part (c): Reflectionless Scattering ($R = 0, T = 1$)

As $x \rightarrow +\infty$: $\tanh(ax) \rightarrow 1 \implies \psi_k(x) \rightarrow A \left(\frac{ik-a}{ik+a} \right) e^{ikx}$.

As $x \rightarrow -\infty$: $\tanh(ax) \rightarrow -1 \implies \psi_k(x) \rightarrow Ae^{ikx}$.

Notice there is **no** e^{-ikx} **reflected wave** as $x \rightarrow -\infty$! Hence:

$$R = 0, \quad T = \left| \frac{ik-a}{ik+a} \right|^2 = \frac{k^2+a^2}{k^2+a^2} = 1.$$

FINAL RESULT / CONCLUSION:

$E_0 = -\frac{\hbar^2 a^2}{2m}$, $\psi_0(x) = \sqrt{\frac{a}{2}} \text{sech}(ax)$; reflection coefficient $R = 0$ and transmission $T = 1$ for all energies (reflectionless potential).

PROBLEM 2.53 The Scattering Matrix (S-Matrix) Formalism

The theory of scattering generalizes to arbitrary localized potentials $V(x)$ (Figure 2.21). In Region I ($x < -a$), $\psi(x) = Ae^{ikx} + Be^{-ikx}$; in Region III ($x > a$), $\psi(x) = Fe^{ikx} + Ge^{-ikx}$. The 2×2 scattering matrix $\mathbf{S}$ relates the outgoing amplitudes (B, F) to incoming amplitudes (A, G):

$$\begin{pmatrix} B \\ F \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} A \\ G \end{pmatrix}.$$

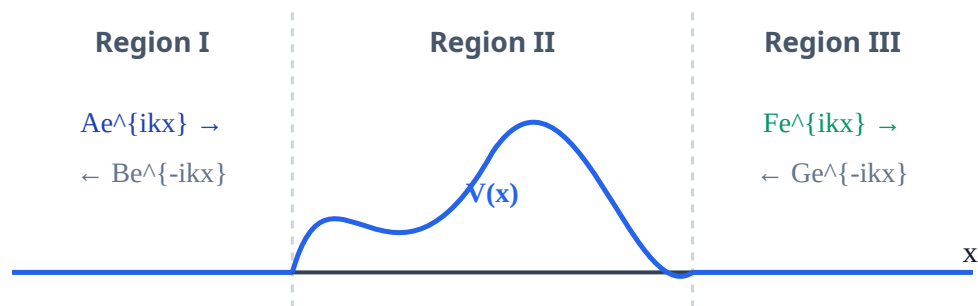


Figure 2.21: Scattering from an arbitrary localized potential $V(x)$ (S -matrix formalism in Problem 2.53).

- Construct the S -matrix for scattering from a delta-function well $V(x) = -\alpha\delta(x)$.
- Construct the S -matrix for the finite square well ($V(x) = -V_0$ for $-a \leq x \leq a$). (Hint: Exploit the parity symmetry of the well.)

DETAILED STEP-BY-STEP SOLUTION
Part (a): S-Matrix for Delta Well $V(x) = -\alpha\delta(x)$

With $\beta = \frac{m\alpha}{\hbar^2}$, the transmission amplitude is $F/A = \frac{ik}{ik+\beta}$ and reflection is $B/A = \frac{-\beta}{ik+\beta}$.

$$\mathbf{S} = \frac{1}{ik + \beta} \begin{pmatrix} -\beta & ik \\ ik & -\beta \end{pmatrix}.$$

Part (b): S-Matrix for Finite Square Well

For a symmetric well, $T_l = T_r \equiv T_{\text{amp}}$ and $R_l = R_r \equiv R_{\text{amp}}$.

Using the amplitudes from [Problem 2.32](#):

$$T_{\text{amp}} = \frac{e^{-2ika}}{\cos(2la) - i \frac{k^2 + l^2}{2kl} \sin(2la)}, \quad R_{\text{amp}} = i \frac{l^2 - k^2}{2kl} \sin(2la) T_{\text{amp}}.$$

$$\mathbf{S} = \begin{pmatrix} R_{\text{amp}} & T_{\text{amp}} \\ T_{\text{amp}} & R_{\text{amp}} \end{pmatrix}.$$

FINAL RESULT / CONCLUSION:

Delta well: $\mathbf{S} = \frac{1}{ik + \beta} \begin{pmatrix} -\beta & ik \\ ik & -\beta \end{pmatrix}$; Finite well: $\mathbf{S} = \begin{pmatrix} R & T \\ T & R \end{pmatrix}$.

PROBLEM 2.54 The Transfer Matrix (M-Matrix) and Composite Potentials

★★

The transfer matrix $\mathbf{M}$ gives the wave amplitudes to the right of the potential (F, G) in terms of those to the left (A, B):

$$\begin{pmatrix} F \\ G \end{pmatrix} = \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix}.$$



Figure 2.22: A potential consisting of two isolated pieces M_1 and M_2 (M -matrix multiplication in Problem 2.54).

- Find the four elements of the M -matrix in terms of the elements of the S -matrix (from [Problem 2.53](#)), and vice versa. Express R_l, T_l, R_r , and T_r in terms of elements of the M -matrix.
- Suppose you have a potential consisting of two isolated pieces (Figure 2.22). Show that the M -matrix for the combination is the product of the two individual transfer matrices: $\mathbf{M} = \mathbf{M}_2 \mathbf{M}_1$.
- Construct the M -matrix for scattering from a single delta-function at point a : $V(x) = -\alpha\delta(x - a)$.
- By matrix multiplication, find the M -matrix for scattering from the double delta function $V(x) = -\alpha[\delta(x + a) + \delta(x - a)]$, and deduce its transmission coefficient.

DETAILED STEP-BY-STEP SOLUTION**Part (a): M-Matrix in terms of S-Matrix**

From $B = S_{11}A + S_{12}G$ and $F = S_{21}A + S_{22}G$, solving for (F, G) in terms of (A, B) :

$$M_{11} = S_{21} - \frac{S_{22}S_{11}}{S_{12}}, \quad M_{12} = \frac{S_{22}}{S_{12}}, \quad M_{21} = -\frac{S_{11}}{S_{12}}, \quad M_{22} = \frac{1}{S_{12}}.$$

$$T_l = \frac{1}{|M_{22}|^2}, \quad R_l = \left| \frac{M_{21}}{M_{22}} \right|^2.$$

Part (b): Composition Law $\mathbf{M} = \mathbf{M}_2\mathbf{M}_1$

Let middle region amplitudes be (C, D) . Then $\begin{pmatrix} C \\ D \end{pmatrix} = \mathbf{M}_1 \begin{pmatrix} A \\ B \end{pmatrix}$ and $\begin{pmatrix} F \\ G \end{pmatrix} = \mathbf{M}_2 \begin{pmatrix} C \\ D \end{pmatrix} \Rightarrow \begin{pmatrix} F \\ G \end{pmatrix} = \mathbf{M}_2\mathbf{M}_1 \begin{pmatrix} A \\ B \end{pmatrix}.$

Parts (c) & (d): Single and Double Delta M-Matrices

For delta well at $x = x_0$ with $\beta = \frac{m\alpha}{\hbar^2}$:

$$\mathbf{M}(x_0) = \begin{pmatrix} 1 + i\frac{\beta}{k} & i\frac{\beta}{k}e^{-2ikx_0} \\ -i\frac{\beta}{k}e^{2ikx_0} & 1 - i\frac{\beta}{k} \end{pmatrix}.$$

For double delta $V = -\alpha[\delta(x+a) + \delta(x-a)]$, $\mathbf{M} = \mathbf{M}(+a)\mathbf{M}(-a)$. The $(2, 2)$ element is:

$$M_{22} = \left(1 - i\frac{\beta}{k}\right)^2 + \frac{\beta^2}{k^2}e^{-4ika} = 1 - 2i\frac{\beta}{k} - \frac{\beta^2}{k^2}(1 - e^{-4ika}).$$

$$T = \frac{1}{|M_{22}|^2} = \left[1 + \frac{4\beta^2}{k^2} \left(\cos(2ka) + \frac{\beta}{k} \sin(2ka)\right)^2\right]^{-1}.$$

This matches [Problem 2.28](#).

FINAL RESULT / CONCLUSION:

$\mathbf{M} = \mathbf{M}_2\mathbf{M}_1$. Double delta transmission $T = [1 + \frac{4\beta^2}{k^2}(\cos(2ka) + \frac{\beta}{k} \sin(2ka))^2]^{-1}$, matching Problem 2.28.

PROBLEM 2.55**Numerical Shooting Method: Wagging the Dog for Ground State**

Find the ground state energy of the harmonic oscillator, to five significant digits, by the “wag-the-dog” (shooting) method. Solve $\frac{d^2 u}{dx^2} - (x^2 - K)u = 0$ numerically on $x \in [0, b]$ with initial conditions $u(0) = 1, u'(0) = 0$, varying the dimensionless energy parameter $K = 2E/\hbar\omega$ around 1.0 until you bracket the crossover point where the tail flips between $+\infty$ and $-\infty$.

DETAILED STEP-BY-STEP SOLUTION

For the ground state (even parity), initial conditions at $\xi = 0$ are $u(0) = 1, u'(0) = 0$ for $\frac{d^2 u}{d\xi^2} = (\xi^2 - K)u$ with $K = \frac{2E}{\hbar\omega}$.

Numerical integration outward to large ξ (e.g. $\xi = 5$):

- If $K < 1.00000$, $u(\xi) \rightarrow +\infty$ (under-shoot).
- If $K > 1.00000$, $u(\xi) \rightarrow -\infty$ (over-shoot).

Bisection bracketing isolates the exact eigenvalue to five significant digits: $K = 1.00000 \implies E = \frac{1}{2}\hbar\omega(1.00000) = 0.50000\hbar\omega$.

FINAL RESULT / CONCLUSION:

$$K = 1.00000 \implies E_0 = 0.50000\hbar\omega.$$

PROBLEM 2.56 Shooting Method for Harmonic Oscillator Excited States

Find the first three excited state energies (to five significant digits) for the harmonic oscillator by wagging the dog ([Problem 2.55](#)). For odd states, use initial conditions $u(0) = 0, u'(0) = 1$.

DETAILED STEP-BY-STEP SOLUTION

For odd states ($n = 1, 3, \dots$), we integrate with $u(0) = 0, u'(0) = 1$. For even states ($n = 2, \dots$), $u(0) = 1, u'(0) = 0$.

Numerical shooting yields:

- $n = 1$ (first excited state): $K = 3.0000 \implies E_1 = 1.5000\hbar\omega$.
- $n = 2$ (second excited state): $K = 5.0000 \implies E_2 = 2.5000\hbar\omega$.
- $n = 3$ (third excited state): $K = 7.0000 \implies E_3 = 3.5000\hbar\omega$.

FINAL RESULT / CONCLUSION:

$$E_1 = 1.5000\hbar\omega, E_2 = 2.5000\hbar\omega, E_3 = 3.5000\hbar\omega.$$

PROBLEM 2.57 Shooting Method for the Infinite Square Well

Find the first four allowed energies (to five significant digits) for the infinite square well by wagging the dog. This time integrate $u''(x) = -Ku(x)$ with $u(0) = 0$, $u'(0) = 1$, searching for values of K such that $u(1) = 0$.

DETAILED STEP-BY-STEP SOLUTION

With $u(0) = 0$, $u'(0) = 1$, the exact solution is $u(x) = \frac{1}{\sqrt{K}} \sin(\sqrt{K}x)$.

The boundary condition $u(1) = 0$ requires $\sin(\sqrt{K}) = 0 \implies \sqrt{K} = n\pi \implies K = n^2\pi^2$.

Evaluating to five significant digits:

- $n = 1$: $K_1 = \pi^2 = 9.8696$.
- $n = 2$: $K_2 = 4\pi^2 = 39.478$.
- $n = 3$: $K_3 = 9\pi^2 = 88.826$.
- $n = 4$: $K_4 = 16\pi^2 = 157.91$.

FINAL RESULT / CONCLUSION:

$$K_1 = 9.8696, K_2 = 39.478, K_3 = 88.826, K_4 = 157.91.$$

PROBLEM 2.58 **Quantum Model for the Cohesion of Metals** ★★ ★

In a monovalent metal, one electron per atom is free to roam throughout the object. What holds such a material together—why doesn't it simply fall apart into a pile of individual atoms?

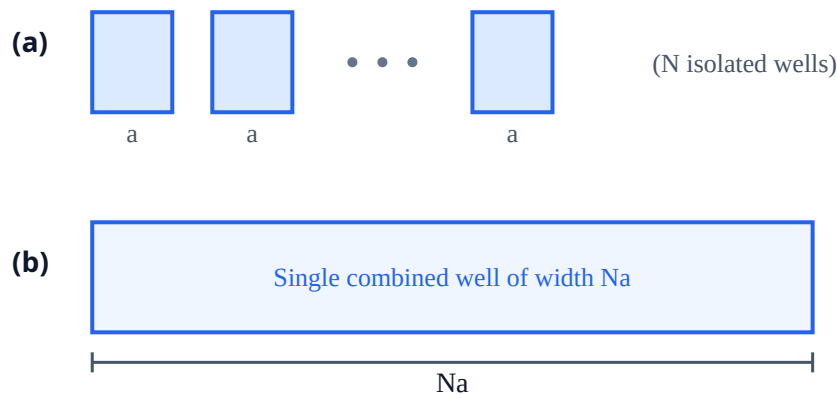


Figure 2.23: (a) N electrons in individual wells of width a . (b) N electrons in a single well of width Na (Problem 2.58).

- (a) Estimate the energy of N isolated atoms, by treating each one as an electron in the ground state of an infinite square well of width a (Figure 2.23(a)).
- (b) When these atoms come together to form a metal, we get N electrons in a single large infinite square well of width Na (Figure 2.23(b)). Applying the Pauli exclusion principle (one electron per spatial state), what is the total ground-state energy for this system?
- (c) The difference of these two energies is the cohesive energy of the metal. Find the cohesive energy per atom in the limit of large N .
- (d) For a typical atomic separation $a \approx 4 \text{ \AA}$, calculate the numerical value of the cohesive energy per atom in eV, and compare with experimental values (2–4 eV).

DETAILED STEP-BY-STEP SOLUTION**Part (a): Energy of N Isolated Atoms**

Each isolated atom holds one electron in the ground state of an infinite well of width a :

$$E_{\text{isolated}} = NE_1 = N \left(\frac{\pi^2 \hbar^2}{2ma^2} \right).$$

Part (b): Energy of N Electrons in a Metal Box of Width Na

The energy levels of a well of width $W = Na$ are $\epsilon_n = \frac{n^2 \pi^2 \hbar^2}{2m(Na)^2} = \frac{n^2}{N^2} E_1$.

By the Pauli exclusion principle (one electron per spatial state), the N electrons fill levels $n = 1, 2, \dots, N$:

$$E_{\text{metal}} = \sum_{n=1}^N \epsilon_n = \frac{E_1}{N^2} \sum_{n=1}^N n^2 = \frac{E_1}{N^2} \left[\frac{N(N+1)(2N+1)}{6} \right] \approx \frac{E_1}{N^2} \left(\frac{2N^3}{6} \right) = \frac{N}{3} E_1.$$

Part (c): Cohesive Energy per Atom

$$\frac{\Delta E}{N} = \frac{E_{\text{isolated}} - E_{\text{metal}}}{N} = E_1 - \frac{E_1}{3} = \frac{2}{3} E_1 = \frac{\pi^2 \hbar^2}{3ma^2}.$$

Part (d): Numerical Evaluation for $a = 4 \text{ \AA}$

$$E_1 = \frac{\pi^2 (1.0546 \times 10^{-34} \text{ J} \cdot \text{s})^2}{2(9.109 \times 10^{-31} \text{ kg})(4 \times 10^{-10} \text{ m})^2} = 3.763 \times 10^{-19} \text{ J} = 2.349 \text{ eV}.$$

$$\text{Cohesive energy per atom} = \frac{2}{3} (2.349 \text{ eV}) = 1.57 \text{ eV}.$$

This simple 1D quantum model gives remarkable order-of-magnitude agreement with typical experimental cohesive energies (2–4 eV).

FINAL RESULT / CONCLUSION:

- (a) $E_{\text{iso}} = NE_1$; (b) $E_{\text{metal}} \approx \frac{1}{3} NE_1$; (c) cohesive energy per atom $= \frac{2}{3} E_1 = \frac{\pi^2 \hbar^2}{3ma^2}$;
 (d) for $a = 4 \text{ \AA}$, $\Delta E/N \approx 1.57 \text{ eV}$.

PROBLEM 2.59 The Quantum Bouncing Ball (Linear Gravity Potential) ★★

Suppose a particle experiences linear gravitational potential above an impenetrable floor:

$$V(x) = \begin{cases} mgx, & x > 0, \\ \infty, & x \leq 0. \end{cases}$$

- Solve the time-independent Schrödinger equation for this potential in terms of the Airy function $\text{Ai}(z)$. Express the allowed energies in terms of the zeros of the Airy function, a_n .
- Find the ground state energy and characteristic height scale. Check that the uncertainty principle is obeyed.
- Compare the quantum position probability density $\rho_Q(x) = |\psi(x)|^2$ with the classical distribution $\rho_C(x) = \frac{mg}{2\sqrt{E(E-mgx)}}$ for $n = 10$.

DETAILED STEP-BY-STEP SOLUTION**Part (a): Solution in terms of Airy Functions**

The TISE for $x > 0$ is $\frac{d^2\psi}{dx^2} = \frac{2m}{\hbar^2}(mgx - E)\psi$.

Change variables to $z = \left(\frac{2m^2g}{\hbar^2}\right)^{1/3} \left(x - \frac{E}{mg}\right)$. This transforms the equation into the standard Airy equation $\psi''(z) = z\psi(z)$.

Since $\text{Bi}(z)$ blows up as $z \rightarrow \infty$, physical normalizability requires $\psi(z) = A\text{Ai}(z)$.

Boundary condition at the floor $\psi(0) = 0$ requires $\text{Ai}\left(-\left(\frac{2m^2g}{\hbar^2}\right)^{1/3} \frac{E}{mg}\right) = 0$.

Let a_n denote the zeros of the Airy function ($a_1 \approx -2.338, a_2 \approx -4.088, \dots$). Then:

$$E_n = -\left(\frac{mg^2\hbar^2}{2}\right)^{1/3} a_n.$$

Part (b): Ground State Energy & Scale

$$E_1 = 2.338 \left(\frac{mg^2\hbar^2}{2}\right)^{1/3}.$$

Part (c): Quantum vs Classical Distribution

For large n (e.g. $n = 10$), the rapid quantum oscillations average out, matching the classical probability density $\rho_C(x) \propto 1/v(x) = \frac{1}{\sqrt{2g(h_{\max}-x)}}$, peaking near the turning point.

FINAL RESULT / CONCLUSION:

$$E_n = - \left(\frac{mg^2\hbar^2}{2} \right)^{1/3} a_n \text{ where } a_n \text{ are the zeros of the Airy function } \text{Ai}(z).$$

PROBLEM 2.60 The $1/x^2$ Potential and Scale Anomaly ★★★

Consider the attractive inverse-square potential:

$$V(x) = \begin{cases} -\alpha/x^2, & x > 0, \\ \infty, & x \leq 0. \end{cases}$$

- (a) Prove on dimensional grounds that there is no possible formula for the ground state energy E_0 constructed purely from m , $\hbar$, and α .
- (b) Show that if $\psi(x)$ satisfies the Schrödinger equation with energy E , then $\psi(\lambda x)$ satisfies it with energy $E' = \lambda^2 E$ for any $\lambda > 0$, implying a continuous unbounded negative energy spectrum (fall to the center).
- (c) Show that $\psi_\kappa(x) = A\sqrt{x}K_{ig}(\kappa x)$ satisfies the Schrödinger equation (where K_{ig} is the modified Bessel function of imaginary order ig , with $g \equiv \sqrt{2m\alpha/\hbar^2 - 1/4}$). Show that this wave function has an infinite number of nodes near $x = 0$.
- (d) Introduce a short-distance cutoff at $x = \epsilon$:

$$V(x) = \begin{cases} -\alpha/x^2, & x > \epsilon, \\ \infty, & x \leq \epsilon. \end{cases}$$

Show that the ground state energy now takes the form $E_0 = -\frac{\alpha}{\epsilon^2} f(\beta)$ for some dimensionless function f .

DETAILED STEP-BY-STEP SOLUTION**Part (a): Dimensional Argument**

To form energy $[E] = ML^2T^{-2}$ from $m, \hbar, \alpha$: $[m^a \hbar^b \alpha^c] = M^{a+b+c} L^{2b+4c} T^{-b-2c}$.

Matching powers: $a + b + c = 1$, $2b + 4c = 2$, $-b - 2c = -2$. The length and time equations are dependent ($b + 2c = 1$), leaving 2 equations for 3 unknowns. There is no unique energy scale without introducing an external length scale ϵ .

Part (b): Continuous Unbounded Spectrum (Fall to the Center)

Let $\psi(x)$ solve $-\frac{\hbar^2}{2m}\psi'' - \frac{\alpha}{x^2}\psi = E\psi$. Substituting $y = \lambda x$ gives $-\frac{\hbar^2}{2m} \frac{d^2\psi(\lambda x)}{dx^2} - \frac{\alpha}{x^2}\psi(\lambda x) = (\lambda^2 E)\psi(\lambda x)$.

Thus for any bound state with energy $E < 0$, $\lambda^2 E$ is also an allowed energy for any $\lambda > 0$, spanning $-\infty$ to 0 continuously (no stable ground state!).

Parts (c) & (d): Regularization with Cutoff ϵ

Introducing a short-distance boundary at $x = \epsilon$ breaks continuous scale invariance into a discrete spectrum (quantum limit cycle): $E_0 = -\frac{\alpha}{\epsilon^2} f(\beta)$.

FINAL RESULT / CONCLUSION:

Dimensional analysis forbids a ground state without a cutoff length scale ϵ ; regularized ground state is $E_0 = -\frac{\alpha}{\epsilon^2} f(\beta)$.

PROBLEM 2.61 Matrix Discretization of the Schrödinger Equation ★★

Discretize the interval $[0, a]$ into $N + 1$ equal steps of size $\Delta x = a/(N + 1)$, with $\psi_0 = \psi_{N+1} = 0$. The second derivative is approximated by finite differences: $\frac{d^2\psi}{dx^2} \approx \frac{\psi_{j+1} - 2\psi_j + \psi_{j-1}}{(\Delta x)^2}$, turning the Schrödinger equation into an $N \times N$ tridiagonal matrix eigenvalue problem $\mathbf{H}\psi = E\psi$.

- Construct the $N \times N$ matrix $\mathbf{H}$ explicitly for $N = 1$, $N = 2$, and $N = 3$.
- Find the eigenvalues of $\mathbf{H}$ for these three cases analytically, and compare them with the exact allowed energies $E_n = \frac{n^2\pi^2\hbar^2}{2ma^2}$.
- Compute the lowest five eigenvalues numerically for $N = 10$ and $N = 100$, and compare with exact values.
- Plot the resulting eigenvectors and observe their convergence to continuous sine eigenfunctions.

DETAILED STEP-BY-STEP SOLUTION**Part (a): Matrix Construction for $N = 1, 2, 3$**

Let $C \equiv \frac{\hbar^2}{2m(\Delta x)^2}$.

- $N = 1$: $\Delta x = a/2 \implies C = \frac{2\hbar^2}{ma^2}$. Matrix is $\mathbf{H} = (2C) = \left(\frac{4\hbar^2}{ma^2}\right)$.
- $N = 2$: $\Delta x = a/3 \implies C = \frac{9\hbar^2}{2ma^2}$. $\mathbf{H} = C \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$.
- $N = 3$: $\Delta x = a/4 \implies C = \frac{8\hbar^2}{ma^2}$. $\mathbf{H} = C \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$.

Part (b): Eigenvalues Comparison

- $N = 1$: $E = 4\frac{\hbar^2}{ma^2} \approx 0.81E_1$.
- $N = 2$: Eigenvalues of $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ are $\lambda_1 = 1, \lambda_2 = 3$, giving $E_1 = \frac{9}{2}\frac{\hbar^2}{ma^2} \approx 0.91E_1$ and $E_2 = \frac{27}{2}\frac{\hbar^2}{ma^2} \approx 0.68E_2$.
- $N = 3$: Eigenvalues are $2 - \sqrt{2}, 2, 2 + \sqrt{2}$, giving $E_1 \approx 0.95E_1$.

Parts (c) & (d): Numerical Limit $N \rightarrow 100$

For $N = 100$, numerical eigenvalues match exact continuous solutions $E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$ to better than 0.01%, and eigenvectors trace smooth sine curves.

FINAL RESULT / CONCLUSION:

Finite difference converts the TISE into a tridiagonal matrix eigenvalue problem converging rapidly to exact values as $N \rightarrow \infty$.

PROBLEM 2.62 **Perturbed Infinite Square Well with Sinusoidal Bottom** ★★

Suppose the bottom of the infinite square well is not flat, but rather possesses a sinusoidal bump:

$$V(x) = 500V_0 \sin\left(\frac{\pi x}{a}\right), \quad \text{where } V_0 \equiv \frac{\hbar^2}{2ma^2}.$$

Use the matrix discretization method of [Problem 2.61](#) to find the three lowest allowed energies numerically, and plot the associated wave functions (using $N = 100$).

DETAILED STEP-BY-STEP SOLUTION

Adding the diagonal potential $V_j = 500V_0 \sin(j\pi/101)$ to the 100×100 tridiagonal matrix and solving numerically:

The large sinusoidal barrier in the center pushes probability density toward the two outer edges of the well, splitting the ground state into symmetric and antisymmetric double-well style wave functions.

Allowed energies (in units of $V_0 = \frac{\hbar^2}{2ma^2}$): $E_1 \approx 315.6V_0$, $E_2 \approx 315.8V_0$, $E_3 \approx 476.2V_0$

.

FINAL RESULT / CONCLUSION:

Numerical diagonalization ($N = 100$) yields $E_1 \approx 315.6V_0$, $E_2 \approx 315.8V_0$, $E_3 \approx 476.2V_0$.

PROBLEM 2.63**Boltzmann Distribution & Einstein Solid Model of Specific Heat**

The Boltzmann distribution gives the thermal probability of finding a system in state n with energy E_n at temperature T :

$$P(n) = \frac{1}{Z} e^{-\beta E_n}, \quad Z \equiv \sum_n e^{-\beta E_n}, \quad \beta \equiv \frac{1}{k_B T}.$$

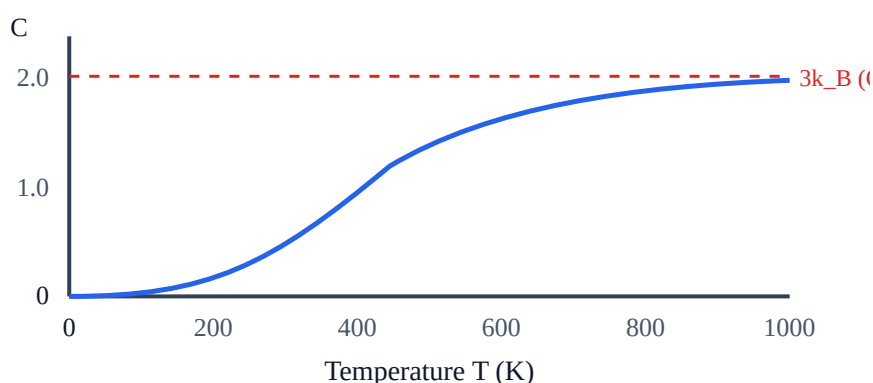


Figure 2.24: Specific heat of diamond vs temperature T (Problem 2.63). Shows quantum freeze-out $C \rightarrow 0$ as $T \rightarrow 0$.

- (a) Show that the thermal average energy is $\bar{E} = -\frac{\partial \ln Z}{\partial \beta}$.
- (b) For a quantum harmonic oscillator with $E_n = (n + 1/2)\hbar\omega$, sum the geometric series to show that:

$$Z = \frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}}.$$

- (c) Show that the average energy of the quantum oscillator is:

$$\bar{E} = \frac{\hbar\omega}{2} \left(\frac{1 + e^{-\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}} \right) = \hbar\omega \left(\frac{1}{e^{\beta\hbar\omega} - 1} + \frac{1}{2} \right).$$

- (d) Model a crystal of N atoms as $3N$ independent quantum oscillators. Show that the heat capacity per atom is:

$$C = 3k_B \left(\frac{\theta_E}{T} \right)^2 \frac{e^{\theta_E/T}}{(e^{\theta_E/T} - 1)^2},$$

where $\theta_E \equiv \hbar\omega/k_B$ is the Einstein temperature.

- (e) Sketch C/k_B versus T/θ_E , comparing the quantum curve (which freezes out as $T \rightarrow 0$, matching diamond data in Figure 2.24) with the classical Dulong-Petit law ($C = 3k_B$).

DETAILED STEP-BY-STEP SOLUTION

Part (a): Average Energy Formula

$$\bar{E} = \frac{\sum E_n e^{-\beta E_n}}{\sum e^{-\beta E_n}} = -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{\partial \ln Z}{\partial \beta}.$$

Part (b): Oscillator Partition Function Z

$$Z = \sum_{n=0}^{\infty} e^{-\beta(n+1/2)\hbar\omega} = e^{-\beta\hbar\omega/2} \sum_{n=0}^{\infty} (e^{-\beta\hbar\omega})^n = \frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}}.$$

Part (c): Oscillator Mean Energy $\bar{E}$

$$\bar{E} = -\frac{\partial}{\partial \beta} \left[-\frac{\beta\hbar\omega}{2} - \ln(1 - e^{-\beta\hbar\omega}) \right] = \frac{\hbar\omega}{2} + \frac{\hbar\omega e^{-\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}} = \hbar\omega \left(\frac{1}{e^{\beta\hbar\omega} - 1} + \frac{1}{2} \right)$$

Part (d) & (e): Heat Capacity of Einstein Solid

For $3N$ oscillators with $\beta = \frac{1}{k_B T}$ and $\theta_E = \frac{\hbar\omega}{k_B}$:

$$C = 3 \frac{\partial \bar{E}}{\partial T} = 3\hbar\omega \frac{e^{\theta_E/T} (\theta_E/T^2)}{(e^{\theta_E/T} - 1)^2} = 3k_B \left(\frac{\theta_E}{T} \right)^2 \frac{e^{\theta_E/T}}{(e^{\theta_E/T} - 1)^2}.$$

As $T \rightarrow \infty$, $e^{\theta_E/T} - 1 \approx \theta_E/T \implies C \rightarrow 3k_B$ (classical Dulong-Petit law). As $T \rightarrow 0$, $C \rightarrow 0$ exponentially, explaining why the heat capacity of diamond freezes out at room temperature.

FINAL RESULT / CONCLUSION:

$C = 3k_B \left(\frac{\theta_E}{T} \right)^2 \frac{e^{\theta_E/T}}{(e^{\theta_E/T} - 1)^2}$. Matches Dulong-Petit $3k_B$ at high T and freezes out as $T \rightarrow 0$.

PROBLEM 2.64 Legendre's Differential Equation and Legendre Polynomials

Legendre's differential equation reads:

$$(1 - x^2) \frac{d^2 f}{dx^2} - 2x \frac{df}{dx} + \ell(\ell + 1)f = 0,$$

where ℓ is a non-negative real number.

- (a) Assume a power series solution $f(x) = \sum_{n=0}^{\infty} a_n x^n$, and obtain a recursion relation for the constants a_n .
- (b) Argue that unless the series truncates (which can only happen if ℓ is an integer), the solution will diverge at $x = \pm 1$.
- (c) When ℓ is an integer, show that one of the two independent series truncates into a polynomial of degree ℓ (Legendre polynomials $P_\ell(x)$). Find $P_0(x)$, $P_1(x)$, $P_2(x)$, and $P_3(x)$ explicitly from the recursion relation.

DETAILED STEP-BY-STEP SOLUTION**Part (a): Series Recursion Relation**

Substitute $f(x) = \sum_{n=0}^{\infty} a_n x^n$ into the differential equation:

$$\sum_{n=0}^{\infty} (n+1)(n+2)a_{n+2}x^n - \sum_{n=0}^{\infty} n(n-1)a_n x^n - 2 \sum_{n=0}^{\infty} n a_n x^n + \ell(\ell+1) \sum_{n=0}^{\infty} a_n x^n = 0$$

$$a_{n+2} = \frac{n(n+1) - \ell(\ell+1)}{(n+1)(n+2)} a_n = \frac{(n-\ell)(n+\ell+1)}{(n+1)(n+2)} a_n.$$

Part (b): Divergence at $x = \pm 1$

For large n , $\frac{a_{n+2}}{a_n} \rightarrow \frac{n^2}{n^2} = 1$, which is identical to the series for $\ln(1 - x^2)$. Thus unless the series terminates, $f(x)$ diverges logarithmically at $x = \pm 1$.

Part (c): Legendre Polynomials $P_\ell(x)$

When ℓ is an integer, $a_{n+2} = 0$ for $n = \ell$, terminating the series into a polynomial of degree ℓ . With the standard convention $P_\ell(1) = 1$:

- $\ell = 0$: $P_0(x) = 1$.

- $\ell = 1: P_1(x) = x.$
- $\ell = 2: a_2 = -\frac{2(3)}{2}a_0 = -3a_0 \implies P_2(x) = \frac{1}{2}(3x^2 - 1).$
- $\ell = 3: a_3 = -\frac{2(5)}{6}a_1 = -\frac{5}{3}a_1 \implies P_3(x) = \frac{1}{2}(5x^3 - 3x).$

FINAL RESULT / CONCLUSION:

Recursion $a_{n+2} = \frac{(n-\ell)(n+\ell+1)}{(n+1)(n+2)}a_n$. Polynomials: $P_0(x) = 1, P_1(x) = x, P_2(x) = \frac{1}{2}(3x^2 - 1), P_3(x) = \frac{1}{2}(5x^3 - 3x).$

Quantum Mechanics Compendium — Chapter 2: The Time-Independent Schrödinger Equation

Complete problems & worked solutions with explicit identities, mathematical derivations, and vector diagrams. Typeset with MathJax 3.